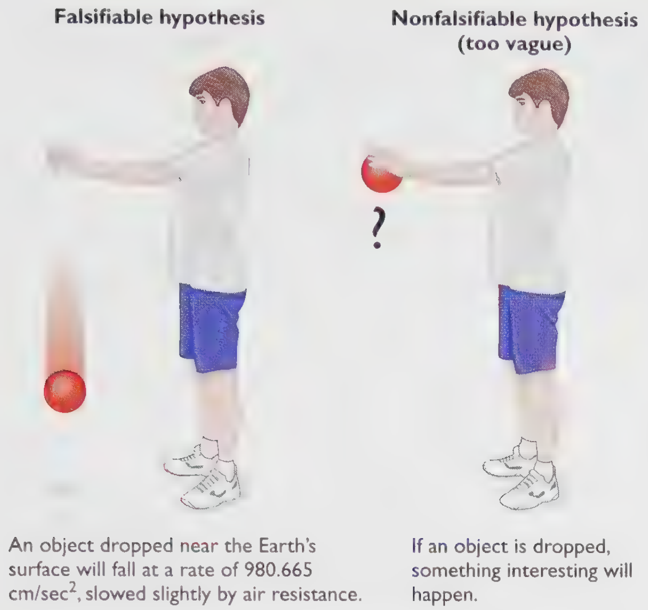


FIGURE 2.2 A good theory makes precise (falsifiable) predictions.

servations that no one had previously expected, then we have reason to be impressed with the theory.

Second, consider what is wrong with this theory: "People who are under too much stress will fail to achieve their full potential." The statement is probably true, but it is too vague to be useful. We do not know how much stress is "too much" for a given individual, and there is no way to measure a person's "full potential." The more precise a theory's predictions, the better. One way of saying this is that a theory should be **falsifiable.** What scientists mean by *falsifiable* is that ***the theory makes sufficiently precise predictions that we can at least imagine evidence that would contradict the theory (if we in fact obtained such evidence).*** Of course, if someone actually obtained evidence that contradicted the predictions, then the theory would be falsifi-*ed*. Falsified is not good. Falsifi-*able* is different; it does not mean that the theory has been shown to be wrong, just that, if it *were* wrong, we could demonstrate it to be wrong. "Too much stress prevents someone from reaching full potential" is not falsifiable, because no conceivable result would contradict it. "Vision occurs after light stimulates the eyes" is falsifiable; vision experienced by an eyeless person or vision in complete darkness would contradict the theory. Therefore, we are impressed that no one has ever found such results.

Now, consider what is wrong with this theory: "I did not wake up on time today, so I must have been kidnapped by aliens." The problem here is that we can easily conceive of a simpler explanation for the fact that someone overslept. Given a choice between an exciting, surprising explanation and a commonplace, boring explanation that fits the same results, scientists favor the commonplace explanation. This preference for simple explanations, known as the *principle of parsimony (literally, stinginess),* is central to scientific thinking, so we shall examine it in detail on the following pages.

CONCEPT CHECK

1. Identify each of the following theories as either falsifiable or not falsifiable:
   a. An object dropped in a vacuum at a low altitude above the Earth will fall with an acceleration of 9.8 m/s^2, where m = meters and s = seconds.
   b. Some people have supernatural powers that science cannot explain.
   c. Children who suffer any kind of frustration during the first year of life will eventually develop emotional difficulties. (Check your answers on page 36.)

SOMETHING TO THINK ABOUT

Is the theory that "there is intelligent life elsewhere in the universe" falsifiable? Is the theory that "there is *no* intelligent life elsewhere in the universe" falsifiable? When it is impractical to test a theory or its converse, should we compromise our usual rules about replicable evidence and falsifiability? ✸

The Principle of Parsimony and Degrees of Open-Mindedness

According to the principle of **parsimony,** *scientists prefer the theory that explains the results using the simplest assumptions.* In other words, scientists prefer a new theory that is consistent with other theories that they have already accepted. A theory that makes radically new assumptions is acceptable only after we have made every attempt to explain the results in a simpler way.

Parsimony is a conservative tendency: It tells us to adhere as much as possible to what we already believe. You might protest: "Shouldn't we remain open-minded to new possibilities?"

Yes, if open-mindedness means a willingness to consider proper evidence, but not if it means the assumption that "anything has as much chance of being true as anything else." That is, the degree of our open-mindedness should depend on the strength of the evidence or logic supporting our current opinion. Consider two examples:

Visitors from outer space. I personally do not believe that visitors from other planets have ever landed on Earth, and I regard the prospect of travel between one solar system and another as unlikely. Still, the technology and the biology of alien life could be very different from ours. If I saw nonhuman pilots stepping out of an odd-looking spacecraft, I would quickly change my opinion about long-distance space travel.

Perpetual motion machines. A "perpetual motion machine" is one that generates more energy than it uses. For centuries, people have attempted and failed to develop such a machine. (Figure 2.3 shows one example.) The U.S. Patent Office is officially closed-minded on this issue, refusing even to consider patent applications for such machines, because a perpetual motion machine violates the second law of thermodynamics. (According to that law, within a closed system, entropy—disorganization—can never decrease. Any work will increase the entropy, and therefore waste some energy.) The second law of thermodynamics is more than just a summary of careful observations; it is a logical necessity, equivalent to stating that a more probable state is more probable than a less probable state. If someone shows you what appears to be a perpetual motion machine, you should not be easily persuaded; look for a hidden battery or other power source. The more extraordinary someone's claim is, the more extraordinary the evidence must be before you should accept their claim.

But what does all this discussion have to do with psychology? Sometimes, people claim spectacular results that would seem to be impossible according to everything else we know or think we know. Although it is only fair to examine the evidence behind such claims, it is also reasonable to maintain a skeptical attitude and to look as hard as possible for a simple, parsimonious explanation of the results. Let us consider two examples.

An Example of Applying Parsimony: Clever Hans, the Amazing Horse

Early in the 20th century, Mr. von Osten, a German mathematics teacher, set out to prove that his horse, Hans, had great intellectual abilities, particularly in arithmetic (Figure 2.4). To teach Hans arithmetic, he first showed him a single object, said "One," and lifted Hans's foot once. Then he raised Hans's foot twice for two objects, and so on. Eventually, when von Osten presented a group of objects, Hans learned to tap his foot by himself, and with practice he managed to tap the correct number of times. With more practice, it was no longer necessary for Hans to see the objects. Von Osten would just call out a number, and Hans would tap the appropriate number of times.

Mr. von Osten moved on to addition and then to subtraction, multiplication, and division. Hans caught on quickly, soon responding with 90–95% accuracy. Then von Osten began touring Germany to exhibit Hans's abilities. He would ask Hans a question, and Hans would tap out the answer. As time passed, Hans's abilities grew, until he could add fractions, convert fractions to decimals or vice versa, do simple algebra, tell time to the minute, and give the values of all German coins. Using a letter-to-number code, he could spell out the names of objects and even identify musical notes such as D or B-flat. (Hans, it seems, had perfect pitch.) He responded correctly even when questions were put to him by persons other than von Osten, in unfamiliar places, with von Osten nowhere in sight.

Given this evidence, many people were ready to assume that Hans had great intellectual prowess. But others sought a more parsimonious explanation. Enter Oskar Pfungst. Pfungst (1911) discovered that Hans could not answer a question correctly unless the questioner had calculated the answer first. Evidently, the horse was somehow getting the answers from the questioner. Next Pfungst learned that, when the experimenter stood in plain sight, Hans's accuracy was 90% or better, but when he could not see the experimenter, his answers were almost always wrong.

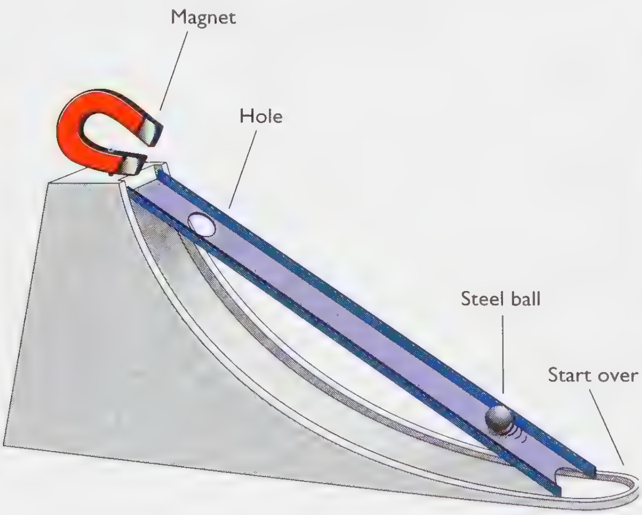


FIGURE 2.3 A proposed "perpetual motion machine": The magnet pulls the metal ball up the inclined plane. When the ball reaches the top, it falls through the hole and returns to its starting point, from which the magnet will again pull it up. Can you see why this device is sure to fail? (See Answer A on page 36.)

FIGURE 2.4 Clever Hans and his owner, Mr. von Osten, demonstrated that the horse could answer complex mathematical questions with great accuracy. The question was "how?" (After Pfungst, 1911, in Fernald, 1984.)

Eventually, Pfungst observed that any questioner who asked Hans a question would lean forward to watch Hans's foot. Hans had simply learned to start tapping whenever someone stood next to his right forefoot and leaned forward. As soon as Hans had given the correct number of taps, the experimenter would give a slight upward jerk of the head and change facial expression in anticipation that this might be the last tap. (Even skeptical scientists who tested Hans did this involuntarily.) Hans simply continued tapping until he received that cue.

In short, Hans was indeed a clever horse. However, his feats could be explained in simple terms. We now prefer the explanation in terms of facial expressions because it is more parsimonious.

SOMETHING TO THINK ABOUT

If Clever Hans had died before Pfungst had discovered his secret, we would never have known for sure how the horse was making his "calculations." Would we be obliged to believe forever that this one horse could understand spoken language and solve complex mathematical problems? How could we have evaluated such a hypothesis years later? (Hint: Would we have needed to discover how Hans answered the questions? Or would it be enough just to determine how he *could* have answered them?) ✷

Master magician Lance Burton can make people and animals seem to suddenly appear, disappear, float in the air, or do other things that we know are impossible. Even if we don't know how he accomplishes these feats, we take it for granted that they are magic tricks, based on methods of misleading the audience. Other performers claim their amazing results depend on psychic powers. A more parsimonious explanation is that their feats, like Burton's, depend on misleading the audience.

Another Example of Applying Parsimony: Extrasensory Perception

One highly controversial topic in psychology is **extrasensory perception.** Supporters of extrasensory perception (ESP) ***claim that certain people can acquire information without using any sense organ and without receiving any form of physical energy.*** These people claim, for instance, that a person gifted with ESP can identify another person's thoughts (via telepathy) even when the two are separated by a thick lead barrier that would block the transmission of almost any form of energy. They also claim that people with telepathic powers can identify thoughts just as accurately from a distance of a thousand kilometers as from an adjacent room, in apparent violation of the inverse-square law of physics.

Some ESP supporters also claim that certain people can perceive inanimate objects that are hidden from sight (clairvoyance), predict the future (precognition), and influence such physical events as a roll of dice by sheer mental concentration (psychokinesis). In other words, they claim it is possible to gain information or to influence physical events without receiving or transmitting any physical energy. A conclusive demonstration of any of these claims would require us not only to overhaul some major concepts in psychology, but also to discard some of the most fundamental tenets of physics.

What evidence is there for ESP?

Anecdotes

One kind of evidence consists of anecdotes—people's reports of isolated events. Someone has a dream or a hunch that comes true or says something and someone else says, "I was just thinking exactly the same thing!" Such experiences may seem impressive when they occur, but they are meaningless as scientific evidence for several reasons: First, consider the possibility of coincidence. Of all the hunches and dreams that people have, some are bound to come true eventually by pure chance. Second, people tend to remember and talk about the hunches and dreams that *do* come true and to forget those that do not. They hardly ever say, "Strangest thing! I had a dream, but then nothing like it actually happened!" Third, people tend to exaggerate the coincidences that occur, both in their own memories and in the retelling. We could evaluate anecdotal evidence only if people recorded their hunches and dreams before the predicted events and then determined how many unlikely predictions actually came to pass.

You may have heard of the "prophet Nostradamus," a 16th-century French writer who allegedly predicted many events in later centuries. Figure 2.5 presents four samples of his writings. No one knows what his predictions mean until after the predicted events happen. After something happens, people imaginatively reinterpret his writings to fit the event. (His predictions are not falsifiable.)

1. The great man will be struck down in the day by a thunderbolt. An evil deed, foretold by the bearer of a petition. According to the prediction another falls at night time. Conflict at Reims, London, and pestilence in Tuscany.
2. When the fish that travels over both land and sea is cast up on to the shore by a great wave, its shape foreign, smooth, and frightful. From the sea the enemies soon reach the walls.
3. The bird of prey flying to the left, before battle is joined with the French, he makes preparations. Some will regard him as good, others bad or uncertain. The weaker party will regard him as a good omen.
4. Shortly afterwards, not a very long interval, a great tumult will be raised by land and sea. The naval battles will be greater than ever. Fires, creatures which will make more tumult.

FIGURE 2.5 According to the followers of Nostradamus, each of these statements is a specific prophecy of a 20th-century event (Cheetham, 1973). Can you figure out what the prophecies mean? Compare your answers to Answer B on page 36. The prophecies of Nostradamus are so vague that no one knows what they mean until after the predicted event. Consequently, they are not really predictions and also not testable.

CONCEPT CHECK

2. How could someone scientifically evaluate the accuracy of Nostradamus's predictions? (Check your answer on page 36.)

Professional Psychics Several stage performers claim to read other people's minds and to perform other psychic feats. Two of the most famous are Uri Geller and the Amazing Kreskin. Actually, Kreskin has consistently denied doing anything supernatural; he prefers to talk of his "extremely sensitive," rather than "extrasensory" perception (Kreskin, 1991). Still, part of Kreskin's success as a performer comes from allowing people to believe he has mental powers that defy explanation, and his performances are similar to those of people who do call themselves psychics.

TRY IT YOURSELF

After carefully observing Geller, Kreskin, and others, David Marks and Richard Kammann (1980) concluded that these performers exhibited the same kinds of deception commonly employed in magic shows. For example, Kreskin sometimes begins his act by asking the audience to read his mind. Let's try to duplicate this trick right now: Try to read my mind. I am thinking of a number between 1 and 50. Both digits are odd numbers, but they are not the same. That is, it could be 15 but it could not be 11. (These are the instructions Kreskin gives.) Have you chosen a number? Please do.

All right, my number was 37. Did you think of 37? If not, how about 35? You see, I started to think 35 and then changed my mind, so you might have got 35.

Probably about half of my readers successfully "read my mind." If you were one of them, are you impressed? Don't be. There are not many numbers that you could have chosen. The first digit had to be 1 or 3, and the second had to be 1, 3, 5, 7, or 9. You had to eliminate 11 and 33 because both digits are the same, and you probably eliminated 15 because I cited it as a possible example. That leaves only seven possibilities. Most people like to stay far away from the example given and tend to avoid the highest and lowest possible choices. That leaves 37 as the most likely choice and 35 as the second most likely.

Second act: Kreskin asks the audience to write down something they are thinking about while he walks along the aisles talking. Then, back on stage, he "reads people's minds." He might say something like, "Someone is thinking about their mother . . . " In any large crowd, someone is bound to stand up and shout, "Yes, that's me, you read my mind!" On occasion he describes something that someone has written out in great detail. That person generally turns out to be someone sitting along the aisle where Kreskin was walking.

After a variety of other tricks (see Marks & Kammann, 1980), Kreskin goes backstage while the local mayor or some other dignitary hides Kreskin's paycheck somewhere in the audience. Then Kreskin comes back, walks up and down the aisles and across the rows, and eventually shouts, "The check is here!" The rule is that if he guesses wrong, then he does not get paid. (He hardly ever misses.)

How does he do this trick? Think for a moment before reading on. Very simply, it is a Clever Hans trick. Kreskin studies people's faces. Most of the people are silently cheering for him to find the check. Their facial expression changes subtly as he comes close to the check and then moves away. In effect, they are saying, "Now you're getting closer" and "Now you're moving away." At last he closes in on the check. That is, Kreskin has trained himself to use his senses very well; he does not need an "extra" sense.

We can also explain the performances of many other stage performers in terms of simple tricks and illusions. Of course, someone always objects, "Well, maybe so. But there's this other guy you haven't investigated yet. Maybe he really does possess psychic powers." Until there is solid evidence to the contrary, it is simpler (more parsimonious) to assume that other performers are also using illusion and deception.

Experiments

Because stage performances and anecdotal events always take place under uncontrolled conditions, we cannot determine the probability of coincidence or the possibility of deception. Laboratory experiments provide the only evidence about ESP worth serious consideration.

For example, consider the *ganzfeld* procedure (from German words meaning "the entire field"). A "sender" is given a photo or film, selected at random from four possibilities, and a "receiver" in another room is asked to describe the sender's thoughts and images. Typically, the receiver wears half Ping-Pong balls over the eyes and listens to static noise over earphones to minimize normal stimuli that might overpower the presumably weaker extrasensory stimuli (Figure 2.6). Later, a judge examines a transcript of what the receiver said and compares it to the four photos or films, determining which one it matches most closely. On the average, it should match the target about one out of four times. If a subject "hits" more often than one in four, we can calculate the probability of accidentally doing that well. (ESP researchers, or parapsychologists, use a variety of other experimental procedures, but in each case the goal is to determine whether someone can gain more information than could be explained by chance without using their senses.)

We can summarize the results of ESP experiments as follows: Over the decades we have gone through many cycles. First, ESP researchers announce that they have good evidence for a given phenomenon. Then, critics find flaws in the procedure or report failures to replicate. Then ESP researchers admit that the previous research was inconclusive but claim that they now have a better procedure and better evidence (Hyman, 1994). Currently, ESP researchers seem most enthusiastic about the ganzfeld method described above. According to one review, six of the ten laboratories that have used this method have reported positive results (Bem & Honorton, 1994); believers claim that researchers have finally found a replicable experiment.

Perhaps, and certainly the best way to test this claim is for more investigators to conduct more research. However, most psychologists remain skeptical—I would say justifiably so. Why? First, they are aware of the long history of flawed procedures, weak evidence, and nonreplicable results in this field of study. With the *ganzfeld* procedures, half of the published research has come from just two laboratories, and one of those laboratories has been strongly criticized for using faulty research methods (Blackmore, 1994). Under the circumstances, it would be premature (at least) to call the latest results "replicable."

Second, psychologists generally look for the most parsimonious explanation for any phenomenon. If someone claims to have a horse that does mathematics or a person claims to be able to read the mind of a person in another room, we should search thoroughly for a simpler explanation and adopt a radically new explanation only if the evidence absolutely and consistently forces us to do so. (Even then, we cannot accept a new explanation until someone specifies it clearly. Saying that some result demonstrates "an amazing ability that science cannot explain" does not constitute an explanation.)

FIGURE 2.6 In the *ganzfeld* procedure, a "receiver," who is deprived of most normal sensory information, tries to describe the photo or film that a "sender" is examining.

THE MESSAGE
Scientific Thinking in Psychology

What have we learned about science in general? Science does not deal with proof or certainty. All scientific conclusions are tentative and are subject to revision. The history of any scientific field contains examples of theories that were once widely accepted and were later revised. Nevertheless, this tentativeness does not imply a willingness to abandon well-established theories in the face of any apparently contradictory evidence.

Scientists always prefer the most parsimonious theory. They abandon accepted theories and assumptions only when better theories and assumptions become available. Scientists closely scrutinize any claim that violates the rule of parsimony. Before they will accept any such claim, they insist that it be supported by replicable experiments that rule out simpler explanations.

SUMMARY

* *Scientific approach to psychology*. Although psychology does not possess the same wealth of knowledge as other sciences, it shares with other scientific fields a commitment to scientific methods, including a set of criteria for evaluating theories. (page 29)
* *Steps in a scientific study*. A scientific study goes through the following sequence of steps: hypothesis, methods, results, interpretation. Because almost any study is subject to more than one possible interpretation, we base our conclusions on a pattern of results from many studies. The results of a given study are taken seriously only if other investigators can replicate them. (page 29)
* *Criteria for evaluating theories*. A good theory agrees with observations and leads to correct predictions of new information. It leads to predictions precise enough that we can state what results would or would not contradict them. All else being equal, scientists prefer the theory that relies on simpler assumptions. (page 30)
* *Skepticism about extrasensory perception*. Claims of the existence of extrasensory perception are scrutinized very cautiously, because the evidence reported so far has been unreplicable and because the scientific approach includes a search for parsimonious explanations. (page 33)

Suggestion for Further Reading

Carey, S. S. (1994). *A beginner's guide to scientific method*. Belmont, CA: Wadsworth. A description of scientific methods, particularly in their application to extraordinary (nonparsimonious) claims.

Terms

hypothesis a testable prediction of what will happen under certain conditions (page 29)

replicable result a result that can be repeated (at least approximately) by any competent investigator who follows the same procedures used in the original study (page 30)

meta-analysis a procedure that combines the results of many studies and analyzes them as though they were all one very large study (page 30)

theory a comprehensive explanation of observable events (page 30)

falsifiable (with reference to a theory) making sufficiently precise predictions that we can at least imagine evidence that would contradict the theory (if anyone had obtained such evidence) (page 31)

parsimony (literally, stinginess) scientists' preference for the theory that explains the results using the simplest assumptions (page 31)

extrasensory perception (ESP) the alleged ability of certain people to acquire information without using any sense organ and without receiving any form of energy (page 33)

Answers to Concept Checks

1. **a.** Falsifiable. Anyone who found an object that had fallen with a different rate of acceleration could contradict the theory, but no one has ever found such an object. This is a good theory because it is *falsifiable* (in principle) and because it has never been *falsified* (by actual data). **b.** Not falsifiable. Even if researchers could scientifically explain every action of every person they had ever tested, the possibility would remain that some untested people have unexplainable powers. (Can you restate the theory so that it is falsifiable?) **c.** Probably not falsifiable, unless someone can specify a reliable method to determine whether a child has suffered "frustration" and a reliable way to determine whether someone has emotional difficulties. As stated, the theory is too vague to be scientifically useful. (page 31)

2. To evaluate Nostradamus's predictions, we would need to ask someone to tell us precisely what his predictions mean before the events they supposedly predict had transpired. Then, we would ask someone else to estimate the likelihood of those events. Eventually, we would compare the accuracy of the predictions to the advance estimates of their probability. That is, we should be impressed with seemingly correct predictions only if our observers had rated these events "unlikely" before they occurred. (page 33)

Answers to Other Questions in the Text

A. Any magnet strong enough to pull the metal ball up the inclined plane would not release the ball when it reached the hole at the top. (page 32)

B. The prophecies of Nostradamus (see page 34), as interpreted by Cheetham (1973), refer to the following: (1) the assassinations of John F. Kennedy and Robert F. Kennedy, (2) Polaris ballistic missiles shot from submarines, (3) Hitler's invasion of France, and (4) World War II.

Web Resources

Center for Social Research Methods
trochim.human.cornell.edu/
Bill Trochim of Cornell University offers Knowledge Base, an interactive on-line textbook for an introductory course in research methods. The site includes links to other Web sites devoted to social research and research methods; an on-line statistical advisor to help you select the appropriate statistical test for your data; common research design exercises; several full-length research papers; and more for the budding researcher.

MODULE 2.2

Methods of Investigation in Psychology

How do psychological researchers study processes that are difficult to define?

How do they design their research, and what special problems can arise?

How do researchers confront the ethical problems of conducting research using both human and nonhuman species?

A radio talk show once featured two psychologists as guests. The first argued that day care was bad for children, because she had seen in her clinical practice many clients who had been left in day care as children and grew up to be sadly disturbed adults. The second psychologist, researcher Sandra Scarr (1997), pointed out that the clinician had no way of knowing about the children left in day care who had become healthy, well-adjusted adults. (Such people have no reason to consult a therapist.) Scarr then described the eight best research studies on this question, which included thousands of children in four countries and found no evidence that day care, even with multiple caregivers, produced any long-term emotional consequences.

What conclusion would you draw? I hope you would conclude that day care is okay for children. To Scarr's dismay, however, the people who called in to the program seemed just as convinced by the anecdotes of disturbed people as by the extensive research studies, and several callers described anecdotes of their own.

Psychology, like any other field, can make progress only by distinguishing between good evidence and weak evidence. Furthermore, psychological research can be tricky; sometimes even a careful investigation runs into unexpected difficulties. In this module we shall consider some of the special problems encountered when applying scientific methods to psychological phenomena.

General Principles for Conducting Psychological Research

The primary goal of this module is not to prepare you to conduct psychological research, although I hope that at least a few readers will eventually do just that. The primary goal is to prepare you to be an intelligent interpreter of psychological research. When you hear about a new study in psychology or a related field, you should be able to ask a few pertinent questions to decide how good the evidence is and what conclusion (if any) it justifies.

Definitions of Psychological Terms

Suppose a physicist asks a student to measure the effect of temperature on the length of an iron bar. The student asks, "What do we really mean by temperature and length?" The physicist might reply, "Don't worry about it. Just measure the length with this ruler and measure the temperature with this thermometer. What I mean by length is the measurement you get with the ruler, and what I mean by temperature is the reading on the thermometer."

We need to use the same strategy for research in psychology. If we wanted to measure the effect of hunger on students' ability to concentrate, we could spend hours attempting to define what hunger and concentration really are, or we could say, "Let's measure hunger by the hours since the last meal and concentration by the length of time that the student continues reading without looking away or doing something else."

By doing so, we would be relying on an **operational definition,** *a definition that specifies the operations (or procedures) used to produce or measure something, a way to give it a numerical value.* An operational definition is not the same as a dictionary definition. You might object that "time since the last meal" is not what you really mean by *hunger.* True, but an operational definition offers a precision that enables one researcher to copy the procedures of another. Later we can worry about what hunger or concentration *really* is.

Suppose that someone wishes to investigate whether children who watch violence on television are likely to behave aggressively themselves. In this case, the investigator needs operational definitions for *televised violence* and *aggressive behavior*. For example, the investigator might define *televised violence* as "the number of acts shown or described in which one person injures another." According to this definition, a 20-minute stalking scene would count as much as a quick attack, and a murder shown on screen would be equivalent to one that the characters just talk about. An unsuccessful attempt to injure someone would not count as violence at all. Also, it is unclear from this definition whether we should count verbal insults. In many ways, this definition might prove to be unsatisfactory, but at least it states how one investigator measures violence. If the results collected on the basis of this definition were confusing, we might try changing the definition.

Similarly, the investigator needs an operational definition of *aggressive behavior.* To define it as "the number of acts of assault or murder committed within 24 hours after watching a particular television program" would be an operational definition but not a very useful one. A better operational definition of *aggressive behavior* specifies less extreme, more likely acts. For example, the experimenter might place a large plastic doll in front of a young child and record how often the child punches it.

Let's take one more example: What is love? Never mind what it *really* is. If we want to measure how it affects another behavior, or how something else affects love, we need an operational definition. One possibility would be "how many hours you are willing to spend with another person who asked you to stay nearby."

CONCEPT CHECK

3. Which of the following is an operational definition of intelligence? **(a)** the ability to comprehend relationships, **(b)** a score on an IQ test, **(c)** the ability to survive in the real world, or **(d)** the product of the cerebral cortex of the brain? What would you propose as an operational definition of friendliness? (Check your answers on page 52.)

Samples of the Population

Researchers generally wish to draw conclusions that apply to a large population, such as all 3-year-olds, or all people with depression, or even all human beings. Because it is not practical to examine everyone in the target population, researchers study a small number of people, a *sample,* and hope that what is true of the sample applies to the whole population.

Finding an appropriate sample is easy when you are studying general mechanisms. For example, early investigators established that the eyes, ears, and other sense organs operate on the same principles for all people (with obvious exceptions such as those with visual or hearing impairments). Indeed, many of these principles are the same even in other animal species. Similarly, many of the principles of learning, memory, hunger, thirst, sleep, and so forth are similar enough among all people that an investigator can explore those principles further with almost any group—students in an introductory psychology class, for example. We refer to ***a group chosen because of its ease of study*** as a **convenience sample.**

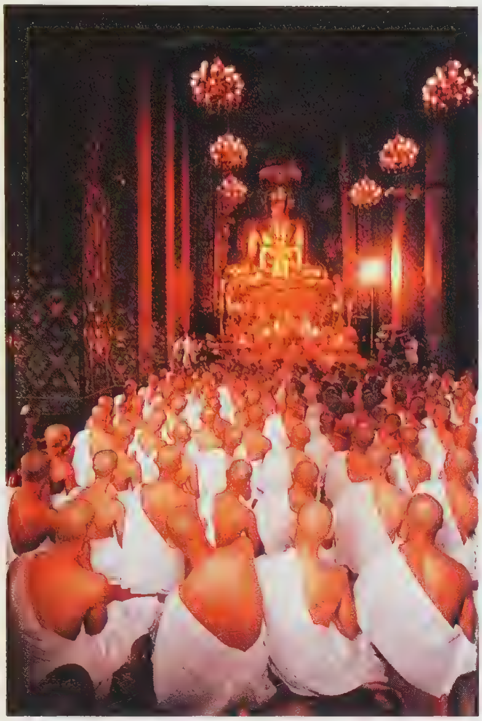
Even with fairly simple behaviors, we can add to our understanding by studying a more diverse sample, however. For example, we can establish general principles about sleep by studying college students, or even a group of laboratory rats, but those studies will not tell us how age differences affect sleep or about the effects of depression on sleep. With many issues, a convenience sample is obviously inappropriate. Imagine that we wanted to determine the frequencies of various political attitudes within a certain country. Obviously, we should not draw generalizations from a study of the students at one college.

To conduct a meaningful study of a behavior that varies significantly among people, we need either a representative sample or a random sample of the relevant population. A **representative sample** *closely resembles the entire population in its percentage of males and females, blacks and whites, young and old, city dwellers and farmers, or whatever other characteristics are likely to affect the results.* To get a representative sample of the people in a given region, an investigator would first determine what percentage of the residents belong to each category and then select people to match those percentages. The disadvantage of this method is that a sample that is representative with regard to one set of characteristics might not be representative of something the investigators did not consider, such as religion or education.

In a **random sample,** *every individual in the population has an equal chance of being selected.* To produce a random sample of city residents, an investigator might select a certain number of city blocks at random from a map of the city, randomly select one house from each of those blocks, and then randomly choose one person from each of those households. A random sample of a thousand people or so is likely to resemble the whole population, but it is difficult to get a truly random sample. We can choose people randomly, but not everyone we choose will agree to participate. Using surveys, we can usually get fairly close to a random sample; with experiments, however, researchers almost always use volunteers. Imagine that I advertise a study on the effects of marijuana on behavior. What kind of people are most likely to volunteer to participate? Probably not a random sample of the population. What if I advertised a study on hypnosis? I would expect to get mostly volunteers who are curious to experience hypnosis, people who might turn out to be easily hypnotized.

Such limitations are not a cause for despair, but certainly a reason to exercise caution. Any researcher who cannot get a random or representative sample should at least state the conclusions cautiously, making it clear that these results may not apply to a broader population.

What if we want to draw generalizations about all humans, not just those in one country? If you imagine trying to get a random or representative sample of all the people on the planet, you will quickly realize the impracticalities. Nevertheless, although we cannot expect to study people from all cultures, it can be very useful to study **cross-cultural samples,** groups of people from at least two cultures, preferably more, and preferably cultures as different as possible. For example, consider some important theoretical issues concerning courtship and marriage: Are certain aspects of male and female behavior predictable consequences of human nature? Or are they all more or less arbitrary customs that a society could readily change? One good way to investigate this question is to compare customs in different societies. Similarly, cross-cultural samples are helpful for studying many other issues: Do people learn facial expressions of emotions? Or is there a fixed relationship between each expression and an emotion? Is a financially prosperous society necessarily a happy society? Do the same psychological disorders occur throughout the world, or are some of them limited to a particular location? And so forth.



A psychological researcher can test generalizations about human behavior by comparing people from different cultures.

Cross-cultural sampling is difficult, however (Matsumoto, 1994). Obvious problems include language barriers and convincing members of another culture to answer personal questions and cooperate with psychological tests that may seem very strange to them. Additional problems arise with behaviors that vary as much within one culture as between cultures. Imagine trying to compare "typical" sexual behaviors between two countries. There may be so much diversity within each country that the comparison between the two countries becomes almost meaningless.

CONCEPT CHECK

4. Suppose I compare the interests and abilities of male and female students at my university. If I find a consistent difference, can I assume that it represents a difference between men in general and women in general? (Obviously the answer is no, or else I would not have asked the question.) Why not? (Check your answer on page 52.)

Experimenter Bias and Blind Studies

When experimenters record their data, sometimes they have trouble separating what they really see from what they expect to see. Here is a class demonstration to illustrate the point: Students in one psychology class were told that they would watch a person engage in a difficult hand-eye coordination task before and after drinking alcohol. That person was in fact drinking only apple juice and performed equally well before and after drinking. Nevertheless, most of the students observing the performance described it as sharply deteriorating (Goldstein, Hopkins, & Strube, 1994). They saw what they had expected to see.

Experimenter bias is *the tendency of an experimenter unintentionally to distort the procedures or results of an experiment based on the expected outcome of the study.* The experimenter does not intend this distortion and may even be trying to avoid it. Imagine that you, as a psychological investigator, are testing the hypothesis that left-handed children are more creative than right-handed children. (I don't know why you would be testing this silly hypothesis, but just suppose you are.) If the results support your hypothesis, you can expect to get your results published and you will be well on your way to becoming a famous psychologist. Now, a left-handed child does something slightly creative. You are not sure whether to count it or not. You want to be fair. You don't want your hypothesis to influence your decision about whether to consider this act creative. Just try to ignore your hypothesis.

To overcome the potential source of error in an investigator's bias, psychologists prefer to use a **blind observer**—that is, *an observer who can record data without knowing what the researcher has predicted.* For example, we might ask someone to record creative acts by a group of children, without any hint that we are interested in the effects of handedness. Because blind observers do not know what hypothesis is being tested, they can record their observations as fairly as possible.

Ideally, the experimenter will conceal the procedure from the subjects as well. For example, suppose experimenters gave one group of children a pill that was supposed to increase their creativity. If those children knew the prediction, maybe they would act more creatively just because

TABLE 2.1 Single-Blind and Double-Blind Studies

Who is aware of which subjects are in which group?

	OBSERVER	SUBJECTS	EXPERIMENTER WHO ORGANIZED THE STUDY
Single-blind	aware	unaware	aware
Single-blind	unaware	aware	aware
Double-blind	unaware	unaware	aware

they knew they were expected to do so. Or maybe the children not taking the pill would pout about not getting it and therefore not do much. The best solution therefore is to give the drug to one group and a **placebo** (*a pill with no known pharmacological effects*) to another group, without telling the children which pill they are taking or what results the experimenter expects. The advantage of this kind of study is that the two groups will not be influenced to behave differently because of their expectations.

A study in which either the observer or the subjects are unaware of which subjects received which treatment is called a **single-blind study** (Table 2.1). *A study in which both the observer and the subjects are unaware* is known as a **double-blind study.** Of course, the experimenter who organized the study would need to keep records of which subjects received which procedure. (A study in which everyone loses track of the procedure is jokingly known as "triple blind.")

Observational Research Designs

The general principles that we just discussed apply to many kinds of research. Psychologists use various methods of investigation, and each has its own advantages and disadvantages. Most research in any field starts with description: What happens, when, and under what circumstances? Astronomers deal almost entirely with observational data; most other scientific fields, including psychology, start with observational data and proceed, when possible, to experiments. Let's start with a description of several kinds of observational studies. Later, we shall consider experiments, which have a much greater ability to illuminate cause-and-effect relationships.

Naturalistic Observations

A **naturalistic observation** is *a careful examination of what many people or nonhuman animals do under more or less natural conditions.* For example, biologist Jane Goodall (1971) spent years observing chimpanzees in the wild, recording their food habits, their social interactions, their gestures, and their whole way of life (Figure 2.7).

Similarly, psychologists sometimes try to observe human behavior "as an outsider." A psychologist might observe what happens when two unacquainted people get on an elevator together: Do they stand close or far apart? Do they speak? Do they look toward each other or away? Does it matter whether the people are two men, two women, or a man and a woman? Do Japanese people act the same as British or Brazilians? A psychologist might also record the behaviors of 6-month-old children, expert crossword puzzle solvers, depressed patients, or any other type of people. After all, the first step toward understanding people is to know in detail what they do.

Case Histories

Psychologists are sometimes interested in rare conditions. For example, some people have amazingly good memories, whereas others have equally amazingly poor memories. People with Williams syndrome have normal or above-average language abilities, but in other regards they are mentally retarded (Bellugi, Wang, & Jernigan, 1994). (Some cannot

FIGURE 2.7 In a naturalistic study, observers record the behavior of people or other species in their natural settings. Here, noted biologist Jane Goodall records her observations on chimpanzees. By patiently staying with the chimps, Goodall gradually won their trust and learned to recognize individual animals. In this manner she was able to add enormously to our understanding of chimpanzees' natural way of life.

even learn to dress themselves.) A psychologist who encounters someone with a rare condition may report a **case history,** *a thorough description of the unusual person.* A case history often relies on naturalistic observation; we distinguish it because it focuses on a single individual.

A case history may include information about the person's medical condition, family background, unusual experiences, current behavior, and a description of the tasks that person can and cannot perform—in short, anything that the investigator thinks might be interesting or important.

A case history is more than just an anecdote. An anecdote is a report of an experience that was not necessarily carefully recorded, such as "I had a hunch and it came true," or "I think I saw Elvis at the pizza parlor." An anecdote is virtually worthless as evidence, because no one can determine whether the person misunderstood or misreported the experience. A case history is potentially replicable by anyone who has the chance to study the same individual or a similar individual again.

A case history is well suited to exploring an unusual behavioral condition; however, the case history does not tell us whether the described individual is typical of others with the same condition. Ideally, a series of case histories can either reveal a pattern or encourage investigators to conduct other kinds of follow-up research.

Surveys

A **survey** is *a study of the prevalence of certain beliefs, attitudes, or behaviors, based on people's responses to specific questions.* A survey is a very common research method. In fact, no matter what your occupation, at some time you will probably conduct a survey yourself, either of your employees, your customers, your students, your neighbors, or fellow members of an organization you have joined. You will also frequently read survey results in the newspaper or hear them reported on television. Thus, you should be aware of some of the difficulties of survey research and understand that survey results can be misleading.

Sampling Getting a truly random or representative sample is a prerequisite for any kind of serious research—and particularly with surveys. Consider what happens with a distorted sample: In 1936, the *Literary Digest* mailed 10 million postcards, asking people their choice for president of the United States. Of the 2 million responses, 57% preferred the Republican candidate, Alfred Landon. Landon was later soundly defeated by the Democratic candidate, Franklin Roosevelt. The problem with this survey was that the *Literary Digest* had selected names from the telephone book and automobile registration lists. In 1936, at the end of the Great Depression, most poor people were Democrats, and very few of them owned telephones or cars.

The Competence of Those Being Interviewed People will often respond to a survey with answers that have not been well thought out. In one 1997 survey, 55% of the respondents said they did not believe in the existence of intelligent life on other planets. However, 82% said they believed the U. S. government was "hiding evidence of intelligent life in space" (Emery, 1997). Granted, it is possible to hide evidence of something that doesn't exist, but more likely some of these people didn't give a lot of thought to their answers.

TRY IT YOURSELF Here's another example: Which of the following programs would you most like to see on television reruns? Rate your choices from highest (1) to lowest (10). (Please fill in your answers, either in the text or on a separate sheet of paper, before continuing to the next paragraph.)

____ Bonanza	____ Jack Benny
____ Car 54	____ My Three Sons
____ Dallas	____ Smothers Brothers
____ Honeymooners	____ Space Doctor
____ I Love Lucy	____ You Bet Your Life

When I conducted this survey at North Carolina State University in 1997, a few of my students refused to answer because they were unfamiliar with most of the listings, but most students ranked all ten programs—including *Space Doctor,* a program that never existed. In fact, 34% of the students ranked this program above average of the 10 programs, and 5% ranked it as their first choice.

Certainly these students did nothing wrong. I asked them to rank all of the programs, and they did. We could only criticize someone who interpreted these results as if they represented informed opinions. This exercise demonstrates that most people will express opinions even when they have no idea what they are talking about.

The Wording of the Questions If a survey includes unclear, ambiguous questions, the results will also be unclear. For example, several years ago, just after basketball star Magic Johnson revealed that he was infected with the HIV virus, one survey asked, "Has the publicity concerning Magic Johnson made you more likely than before to practice safe sex?" A "yes" answer had a reasonably clear meaning. But what did a "no" answer mean? It might mean, "I plan to continue having unsafe sex," or "I was already practicing safe sex, so I did not become more likely to have safe sex." It might even mean, "I was previously having no sex and I expect, unfortunately, to continue not having sex."

Here is another example: Early in 1997, television and other news media briefly mentioned a survey of noted scientists that found that 40% said they believed in God. Some people found this to be an interesting statistic, because it was either higher or lower than they had expected. But unless you know how the question was worded, you won't know what the results mean, and I think some of the write-in answers were far more interesting than the 40% who answered yes. This yes-or-no question was phrased like this:

> I believe in a God in intellectual and affective communication with humankind, i.e., a God to whom one may pray in

> expectation of receiving an answer. By "answer" I mean more than the subjective, psychological effect of prayer.

Some of the scientists objected that they had religious or spiritual beliefs, even though they could not answer yes to the question as it was worded. One wrote, "When backed into a corner, as it were, by questions such as those on the survey I have to come down on the side of nonbelief. . . . This result for me, however (and possibly for others), is an unduly harsh picture. I try frequently to open my mind to an influence of what is good, and the 'subjective and psychological' effects of this can be quite profound, such that I am happy to make contact with the religious tradition by saying that I am praying to God."

In short, for many people, scientists or not, the issue of religious faith is not a simple yes-or-no question (Larson & Witham, 1997). Similarly, any other survey question that forces people to provide a yes-or-no answer may thereby overlook a wide range of interesting shades of opinion.

Surveyor Biases Sometimes a survey is sponsored by an organization that is hoping the results will come out in their favor, and therefore words the questions to encourage the desired outcome. Here is an example: According to a 1993 survey, 92% of high school boys and 98% of high school girls claimed to have been victims of sexual harassment (Shogren, 1993). According to the sponsors of the survey, these results show that sexual harassment has reached "epidemic" proportions. Maybe it has, but the wording of the survey *defined* sexual harassment into an epidemic by specifying that sexual harassment included any of a long list of acts, which ranged from very serious offenses (such as having someone rip your clothes off) to a variety of minor annoyances. For example, if someone wrote sexually explicit graffiti on the rest room wall and you read it and found it offensive, then you could consider yourself sexually harassed. If you tried to make yourself look sexually attractive (as most teenagers do, right?) and then attracted a suggestive stare from someone you *didn't* want to attract, that look would count as sexual harassment.

Sexual harassment is indeed a serious problem, but a survey that lumps the major offenses together with the minor ones fails to provide useful information. Figure 2.8 shows a

Survey on Behalf of the Litterers' Society

1. Prior to reading the enclosed letter, were you aware of how little damage litter does to the environment, or how much money the government wastes in an effort to harass innocent citizens who happen to drop a little bit of litter now and then?
   ____ yes ____ no

2. Do you agree that moderate, responsible littering is one of the rights we should expect in a free society?
   ____ yes ____ no ____ undecided

3. Are you in favor of having the government use your tax dollars to arrest and persecute people whose only "crime" is littering?
   ____ yes ____ no ____ undecided

4. Do you see a need for an educational campaign to inform the public about the good that littering contributes to *natural* recycling?
   ____ yes ____ no ____ undecided

5. Are you outraged that short-sighted do-gooders would pass extremely restrictive, punitive laws against littering, merely to advance their own political careers?
   ____ yes ____ no ____ undecided

6. Do you support the noble attempts of the Litterers' Society to fight against excessive and unnecessary regulations, and will you support the Society with your generous donation?
   ____ Yes! Enclosed is my generous donation! ____ Sorry, I can't afford a contribution at this time.

FIGURE 2.8 An example of how to bias a survey: This imaginary survey for an imaginary society has a style of questions similar to those found in many surveys sponsored by actual political and social organizations. The request for a donation is a reliable clue that the organization is not really seeking your opinion and will probably not even bother to tabulate the results.

fictional survey very similar to those you probably receive occasionally from various organizations. Whenever the questions are written to suggest one "correct" answer, you can safely assume that the organization is much more interested in soliciting money than in knowing your honest opinions.

Correlational Studies

Another type of study is a correlational study. A **correlation** is *a measure of the relationship between two variables, which are both outside the investigator's control.* (A variable is a measurable item that can vary in magnitude. Height and weight are variables; so are years of education and reading speed.) For example, investigators have observed that students who attend class regularly generally receive better test scores than do students who frequently miss class (Cavell & Woehr, 1994). This statement highlights a relationship between two variables—class attendance and test scores. The investigator simply measured these variables, without attempting to influence anything.

A survey can be considered a correlational study if the interviewers compare two or more groups. For example, the interviewers might compare the beliefs of men and women or those of young people and old people. The interviewer could thereby measure a relationship between one variable (sex or age) and another (beliefs).

The Correlation Coefficient

Some correlations are strong; others are weak. For example, we would probably find a strong positive correlation between hours per week spent reading novels and scores on a vocabulary test. We would observe a lower correlation between hours spent reading novels and scores on a chemistry test.

It is often helpful to have a method for measuring the direction and strength of a correlation. The standard method is known as a **correlation coefficient,** *a mathematical estimate of the relationship between two variables. The coefficent can range mathematically from +1 to −1.* A correlation coefficient indicates how accurately we can use a measurement or one variable to predict another. A correlation coefficient of +1, for example, means that, as one variable increases, the other increases also. A correlation coefficient of −1 means that as one variable increases, the other decreases. A correlation of either +1 or −1 would enable us to make perfect predictions of either variable whenever we know the other one. (In real life, psychologists seldom encounter a perfect +1 or −1 correlation coefficient.) The closer the correlation coefficient is to +1 or to −1, the stronger the relationship between the two variables and the more accurately we can use one variable to predict the other. For example, the more people practice a video game, the higher their scores tend to be; this is a positive correlation. The more that people practice golf, the lower their golf scores; this is a negative correlation. (Low golf scores are better.) A negative correlation is just as useful as a positive correlation and can indicate just as strong a relationship.

Figure 2.9 shows hypothetical (fictitious) data demonstrating how grades on a final exam in psychology might correlate with five other variables. (This kind of graph is called a scatterplot; each dot represents the measurements of two variables for one person.) Grades on a psychology final exam correlate very strongly with grades on the previous tests in psychology, less strongly with grades on the French final exam, not at all with the person's weight, and negatively with the amount of time spent watching television and the number of times the student was absent from class. Note that a correlation of +0.9 is almost a straight line ascending; a correlation of −0.9 is close to a straight line descending.

Here are three examples of findings from correlational studies:

- The most crowded areas of a city are generally the most impoverished. (Positive correlation between crowdedness and poverty.)

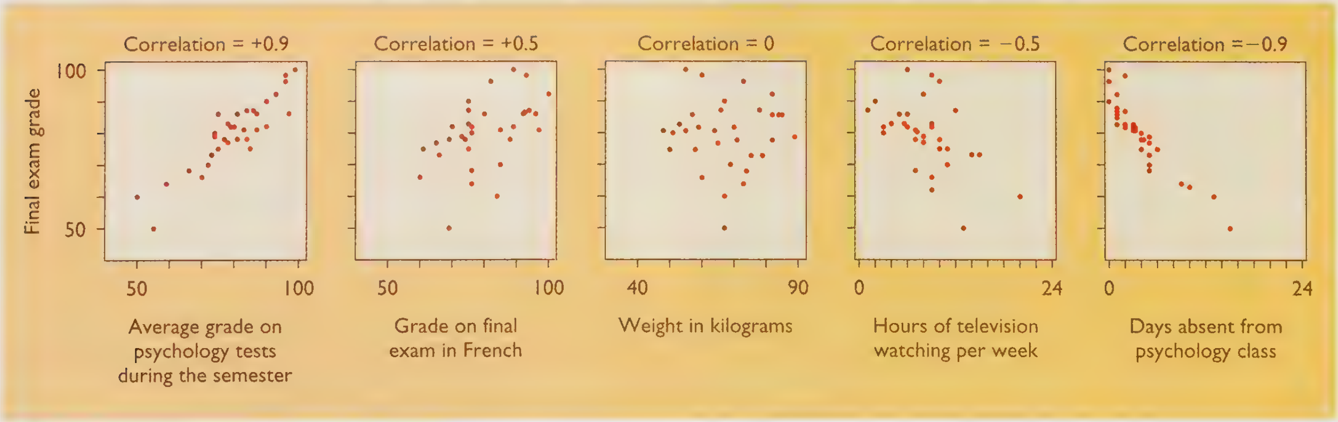


FIGURE 2.9 In a scatterplot, each dot represents data for one person; for example, each point in the center graph tells us one person's weight and that person's grade on the psychology final exam, in this case using hypothetical data. A positive correlation indicates that, as one variable increases, the other generally does also. A negative correlation indicates that, as one variable increases, the other generally decreases. The closer a correlation coefficient is to +1 or −1, the stronger the relationship.

- People's IQ scores have no relationship to their telephone numbers. (Zero correlation between the two.)
- People who trust others are unlikely to cheat other people. (Negative correlation between trusting and cheating.)

CONCEPT CHECK

5. Which indicates a stronger relationship between two variables, a +0.50 correlation or a −0.75 correlation? (Check your answer on page 52.)

Illusory Correlations It is difficult to identify a correlation between two variables solely on the basis of casual observation. For example, years ago, a drug company marketed the drug Bendectin to relieve morning sickness in pregnant women. After a few women who had taken the drug had babies with birth defects, widespread publicity blamed the drug. Thereafter, women who took the drug and had babies with birth defects blamed the drug. Actually, only 2–3% of the women taking the drug had babies with birth defects—the same as the percentage among women not taking the drug! When people expect to see a connection between two events (such as Bendectin and birth defects), they remember the cases that support the connection and disregard the exceptions, thus perceiving an **illusory correlation,** *an apparent relationship based on casual observations of unrelated or weakly related events.* Many social stereotypes are examples of illusory correlations.

As another example of an illusory correlation, consider the widely held belief that a full moon affects human behavior. For hundreds of years, many people have believed that crime and various kinds of mental disturbance are more common under a full moon than at other times. In fact, the term *lunacy* (from the Latin word *luna*, meaning "moon") originally meant mental illness caused by the full moon. Some police officers report that they receive more calls on nights with a full moon, and hospital workers report more emergency cases on such nights. Those reports, however, are based on what people recall rather than on carefully analyzed data. James Rotton and I. W. Kelly (1985) examined all available data relating crime, mental illness, and other phenomena to the phases of the moon. They concluded that the phase of the moon has either no effect at all on human behavior or so little effect that it is almost impossible to measure.

People's expectations and faulty memories produce illusory correlations, such as between the full moon and abnormal behavior.

Why then does this belief persist? We do not know when or how it first arose. (It may have been true many years ago, before the widespread use of artificial lights.) But we can guess why it persists. Suppose, for example, you are working at a hospital. You expect to handle more emergencies on a night with a full moon than at other times. Sooner or later, on a full-moon night, you encounter an unusually high number of accidents, assaults, and suicide attempts. You say, "See? There was a full moon and people just went crazy!" You will remember that night for a long time and disregard all the other full-moon nights when nothing special happened and all the other nights when there was no full moon and yet you were swamped with emergency cases.

Correlation and Causation

A correlational study tells us whether two variables are related to each other and, if so, how strongly. It does not tell us *why* they are related, and a correlation does not justify a cause-and-effect conclusion. For example, there is a very strong positive correlation between the number of books someone owns about chess and how good that person is at playing chess. Does owning chess books cause someone to become a good chess player? Of course not; if it did, I could go buy 50 chess books and suddenly become a great player. Does being a good chess player cause someone to buy chess books? Not necessarily. The real pattern is more complicated: People who start to like chess buy chess books, which help them improve their game. As they get better, they become even more interested, buying more books, reading them, and getting even better at the game. But neither the chess books nor the skill actually causes the other.

"Then, what good is a correlation," you might ask, "if it can't tell us about causation?" There are various answers to this question, but the simplest is that correlations enable us to make useful predictions. If your friend has just challenged you to a game of chess, you can quickly look over your friend's bookshelves and get a fairly good idea of your chances of winning.

Here are two more examples to illustrate why we cannot draw conclusions regarding cause and effect from correlational data (see also Figure 2.10):

- *Unmarried men are more likely than married men to wind up in a mental hospital or prison.* That is, for men, marriage is negatively correlated with mental illness and criminal activity. Does the correlation mean that marriage leads

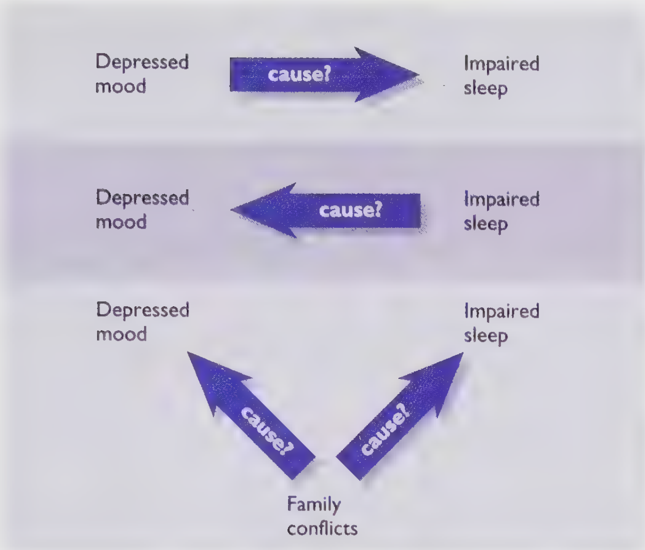


FIGURE 2.10 A strong correlation between depression and impaired sleep does not tell us whether depression interferes with sleep, poor sleep leads to depression, or whether another problem leads to both depression and sleep problems.

to mental health and good social adjustment? Or does it mean that men who are confined to mental hospitals or prisons are unlikely to marry? (Both explanations could, of course, be valid.)

- *People who engage in regular exercise tend to have high self-esteem and tend not to feel depressed.* This correlation could indicate that exercise helps to raise self-esteem and to decrease the chance of depression. It could also mean that nondepressed people with high self-esteem are likely to get some exercise. (Very depressed people seldom do much of anything.) The correlation could also reflect that being young, healthy, and employed increases the chances that a person will find time to exercise and also (independently) improves self-esteem and guards against depression.

To repeat: A correlation does not tell us about causation. To determine causation, an investigator needs to manipulate one of the variables directly, through a research design known as an *experiment*. When an investigator manipulates one variable and then observes corresponding changes in another variable, causation is clear.

CONCEPT CHECK

6. Suppose someone demonstrates a 0.8 correlation between students' reported interest in psychology and their grades on a psychology test. What conclusion can we draw? (Check your answer on page 52.)

Experiments

An **experiment** is *a study in which the investigator manipulates at least one variable while measuring at least one other variable.* The logic behind the simplest possible experiment is as follows: The investigator assembles a suitable sample of people (or animals), divides them randomly into two groups, and then conducts some procedure with one group and not the other. Someone, preferably a blind observer, records the behavior of the two groups. If the behavior of the two groups differs in some consistent way, then the difference is probably the result of the experimental procedure. Table 2.2 contrasts experiments with observational studies.

I shall describe psychological experiments and some of their special difficulties. To illustrate, let's use the example of experiments conducted to determine whether watching violent television programs leads to an increase in aggressive behavior.

Independent Variables and Dependent Variables

An experiment is an attempt to measure the effect of changes in one variable on one or more other variable(s). The **independent variable** is *the item that an experimenter changes or controls—for example, the amount of violent television that the subjects are permitted to watch. The* **dependent variable** is *the item that an ex-*

TABLE 2.2 Comparison of Five Methods of Research

OBSERVATIONAL STUDIES	
Case Study	Detailed description of a single individual; suitable for studying rare conditions.
Naturalistic Observation	Description of behavior under natural conditions; particularly valuable method when it is unethical or impractical to conduct laboratory investigations.
Survey	Study of attitudes, beliefs, or behaviors based on answers to questions.
Correlation	Description of the relationship between two variables that the investigator measures but does not control; determines whether two variables are closely related but does not address questions of cause and effect.
Experiment	Determination of the effect of an independent variable (controlled by the investigator) on the dependent variable (that being measured); the only method that can inform us about cause and effect.

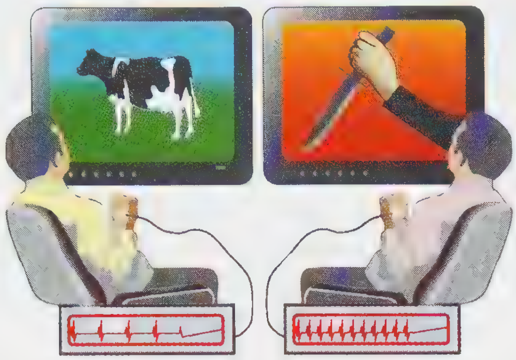


FIGURE 2.11 An experimenter manipulates the independent variable (in this case, the films people watch) so that two or more groups experience different treatments. Then the experimenter measures the dependent variable (in this case, pulse rate) to see how the independent variable affected it.

perimenter measures to determine how it was affected. In our example, the experimenter measures the amount of aggressive behavior that the subjects exhibit. You can think of the independent variable as the influence and the dependent variable as the result (see Figure 2.11).

CONCEPT CHECK

7. An instructor wants to find out whether the frequency of testing in an introductory psychology class has any effect on students' final exam performance. The instructor gives weekly tests in one class, just three tests in a second class, and only a single midterm exam in the third class. All three classes are given the same final exam, and the instructor then compares their performances. Identify the independent variable and the dependent variable. (Check your answers on page 52.)

Experimental Group, Control Group, and Random Assignment

An **experimental group** *receives the treatment that an experiment is designed to test.* In our example, the experimental group would watch televised violence. The **control group** is *a set of individuals treated in the same way as the experimental group except for the procedure that the experiment is designed to test.* People in the control group would watch only nonviolent television programs (Figure 2.12). (The type of television program is the independent variable; the resulting behavior is the dependent variable.)

In principle, this procedure sounds easy, although difficulties can arise in practice. For example, if we are studying a group of teenagers who have a history of violent behavior, it may be difficult to find nonviolent television programs that can hold their attention. As their attention wanders, they might start picking fights with one another, and suddenly the results of the experiment look very odd indeed.

Suppose we conduct a study inviting young people to watch either violent or nonviolent programs, and then we discover that those who watched violent programs act more aggressively. What conclusion could we draw? None, of course. Those who chose to watch violence were probably different from those who chose the nonviolent programs. Any good experiment has **random assignment** of subjects to groups: *The experimenter uses a chance procedure, such as drawing names out of a hat, to make sure that every subject has the same probability as any other subject of being assigned to a given group.*

Consider another example: We set up a rack of cages and put each of several rats in a cage by itself. The rack has five rows of six cages each, numbered from 1 in the upper left-hand corner to 30 in the lower right. Regardless of the procedures we use, we find that the rats with higher cage numbers are more aggressive than those with lower cage numbers. Why?

We might first guess that the difference has to do with location. The rats in the cages with high numbers are farthest from the lights and closest to the floor. They get fed last each day. To test the influence of these factors, we move some of the cages to different positions in the rack, leaving each rat in its own cage. To our surprise, the rats in cages 26–30 are still more aggressive than those in cages 1–5. Why? (How could they possibly know which number is on their cage, and even if they did know, why would they care?)

The answer concerns how rats get assigned to cages. When an investigator buys a shipment of rats, which one goes into cage 1? The one that is easiest to catch! Which ones go into the last few cages? The vicious, ornery little critters that put up the greatest resistance. The rats in the last few cages were already aggressive before they were put into those cages.

The point is that, even when using rats, an experimenter must assign individuals at random to the experimental group and the control group. It would not be random to assign the

Even rats must be assigned to groups randomly.

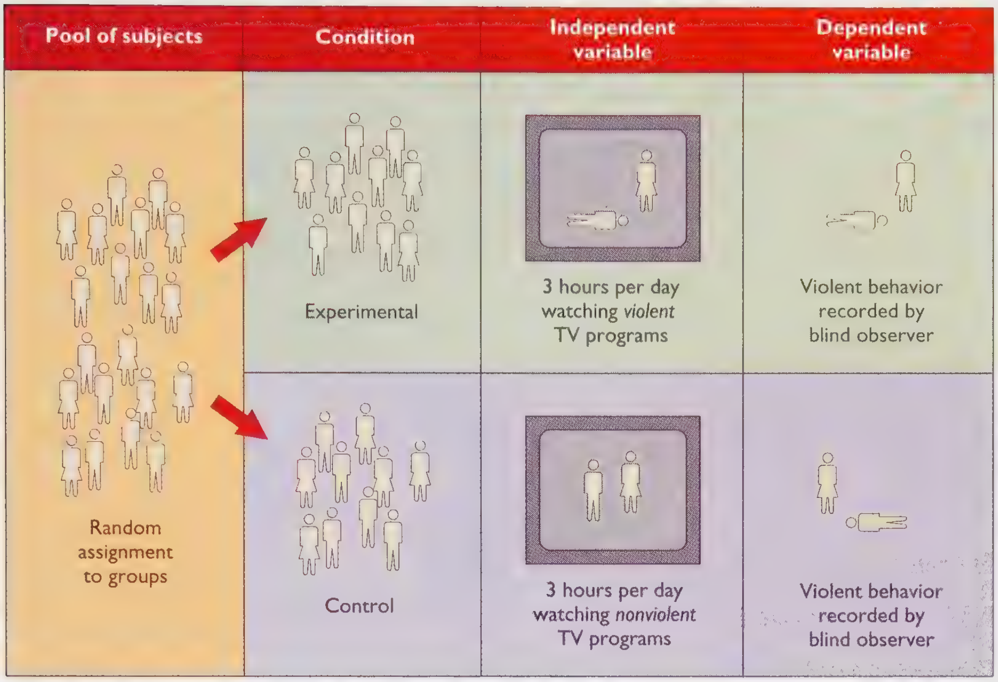


FIGURE 2.12 Once researchers decide on the hypothesis they want to test, they must design the experiment. These procedures test the effects of watching televised violence. An appropriate, accurate method of measurement is essential.

first 15 rats to one group and the second 15 to another group. This advice is even more important with experiments involving humans.

WHAT'S THE EVIDENCE?

Studies of the Effects of Televised Violence on Aggressive Behavior

We have talked in general terms about experiments on the effects of televised violence. Now let's consider some actual examples.

Some of the evidence regarding the effects of televised violence comes from correlational studies. Several such studies have found that people who watch a great deal of televised violence are more likely to engage in aggressive behavior than people who do not (National Institute of Mental Health, 1982). These results are suggestive but inconclusive. They do not tell us whether watching violence leads to aggressive behavior or whether people prone to aggressive behavior like to watch violence on television. To examine a possible cause-and-effect relationship, we must conduct experiments.

Hypothesis Children who watch violent television programs will engage in more acts of aggression than will children who spend the same amount of time watching nonviolent programs.

Method One set of experimenters chose to study male juvenile delinquents in an institution (Parke, Berkowitz, Leyens, West, & Sebastian, 1977). The disadvantage of the study was that the conclusions might apply to only a limited group. The advantage was that the experimenters could control the choice of television programs (the independent variable) much better in a detention center than they could with youngsters living at home.

The boys in the study were randomly assigned to two cottages. Those in one cottage watched violent films on five consecutive nights; those in the other cottage watched nonviolent films. Throughout the study period, blind observers recorded incidents of aggressive behavior by each boy. On the sixth day, each boy was put into an experimental setting where he was given the opportunity at certain times to press a button that, he thought, would deliver an electric shock to another boy. (In fact, no shocks were given.) The experimenters recorded the frequency and intensity of shocks that each boy chose to deliver (the dependent variable).

Results Compared to the boys who had watched nonviolent films, those who had watched the violent films engaged in more acts of aggression and pressed the button to deliver more frequent and more intense electric shocks.

Interpretation At least in this study, watching violent films did lead to increased violence. As with most studies, however, this one had limitations. The boys in this experiment were not representative of boys in general, much less of people in general. Moreover, we cannot assume that we would get similar results with different violent films or different methods of measuring aggressive behavior.

The only way to get around the limitations of a given experiment is to conduct additional experiments, using different samples of people, different films, and different measures of aggressive behavior. Several such experiments have been conducted. Although the results vary considerably, most psychologists have concluded that watching televised violence does indeed increase aggressive behavior, at least temporarily (Comstock & Strasburger, 1990). However, we have little information about the cumulative effects of years of watching violence on television.

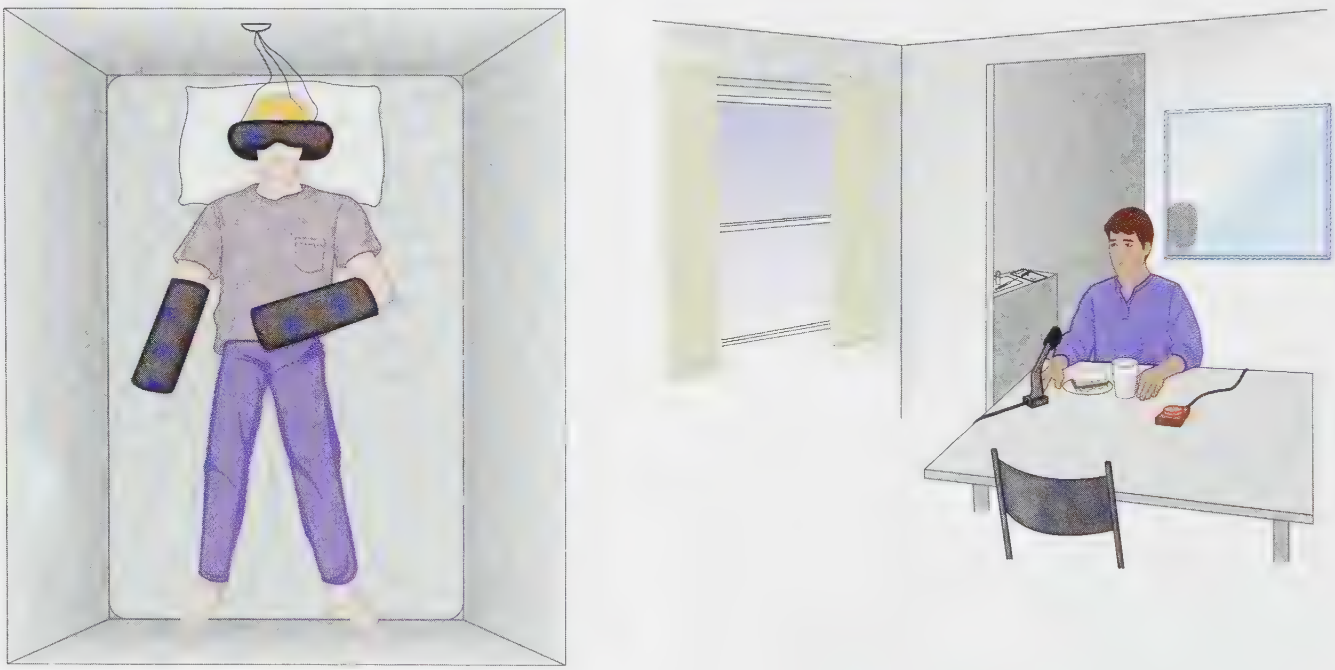
* * *

How Experiments Can Go Wrong: Demand Characteristics

Research on human behavior poses some thorny problems, because people who know they are part of an experiment figure out, or think they have figured out, what is going to happen in the experiment. Their expectations can influence behavior enough to overwhelm the effects of whatever the experimenter is actually doing.

To illustrate: In some well-known studies on sensory deprivation that were popular many years ago, subjects were placed in an apparatus that minimizes vision, hearing, touch, and other forms of sensory stimulation. After several hours, many subjects reported hallucinations, anxiety, and difficulty concentrating. Now, suppose you have heard about these studies, and you agree to participate in an experiment described as a study of "meaning deprivation." The experimenter asks you about your medical history and then asks you to sign a form agreeing not to sue the institution if you have a bad experience. You see an "emergency tray" containing medicines and instruments, which the experimenter assures you is there "just as a precaution." Now you enter an "isolation chamber," which is actually an ordinary room with two chairs, a desk, a window, a mirror, a sandwich, and a glass of water. You are going to be left there for four hours. You are shown a microphone which you can use to report any hallucinations or other distorted experiences and a "panic button" you could press for escape if the discomfort becomes unbearable.

In fact, this is not a study of deprivation at all. Staying in a room by yourself for a few hours should hardly be a traumatic experience. But all the preparations have suggested that terrible things were about to happen, so when this study was actually conducted, several students reported that they were hallucinating "multicolored spots on the wall," or that "the walls of the room are starting to waver," or that "the objects on the desk are becoming animated and moving about" (Figure 2.13). Some complained



a b

FIGURE 2.13 (a) In experiments on sensory deprivation, a person who is deprived of most sensory stimulation becomes disoriented, loses track of time, and reports hallucinations. But do these results partly reflect the person's expectation of having distorted experiences? (b) In one experiment, students were placed in a normal room after undergoing various procedures designed to make them expect a dreadful experience. Many reported hallucinations and distress.

of anxiety, restlessness, difficulty concentrating, and spatial disorientation. One pressed the panic button to demand release (Orne & Scheibe, 1964).

Students in a control group were led to the same room, but they were not shown the "emergency tray," they were not asked to sign a release form, and were given no other indication that they were expected to have any unpleasant experiences. And they, in fact, reported no unusual experiences.

Sensory deprivation may very well have significant effects on behavior. But as this experiment illustrates, we must carefully distinguish between the effects of the independent variable and the effects of what the subjects expect from the experiment. Martin Orne (1969) defined **demand characteristics** as *cues that tell a subject what is expected of him or her and what the experimenter hopes to find.* In a sense, demand characteristics set up *self-fulfilling prophecies*. That is, when designing an experiment, the experimenter has a certain expectation in mind and may then inadvertently convey that expectation to the subjects, thereby influencing them to behave as expected. Many experimenters take elaborate steps to conceal the purpose of the experiment from the subjects to eliminate demand characteristics. A double-blind study serves the purpose: If two groups share the same expectations but behave differently because of the treatment they receive, then the differences in behavior are presumably not the result of their expectations.

CONCEPT CHECK

8. Which of the following would an experimenter try to minimize or avoid?
   a. falsifiability
   b. independent variables
   c. dependent variables
   d. blind observers
   e. demand characteristics
   (Check your answers on page 52.)

Ethical Considerations in Experimentation

In any experiment, psychologists manipulate a variable to determine how it affects behavior. Perhaps the concept of someone trying to alter your behavior sounds objectionable. If so, consider that every time you talk to other people you are trying to alter their behavior at least slightly. Most experiments in psychology produce effects that are no more disruptive than the effects of a conversation.

Still, some experiments do raise ethical issues. Psychologists are seriously concerned about the ethical issues that arise both in the experiments they conduct with humans and in those they conduct with animals.

Ethical Concerns in Experiments on Humans

Earlier in this chapter, I discussed experiments on the effects of televised violence. If psychologists believed that watching violent programs on television would really transform viewers into murderers, then it would be unethical for them to conduct an experiment to find out for sure. It would also be unethical to perform any experimental procedure likely to cause people significant pain or embarrassment or to exert any long-lasting, undesirable effects on their lives.

One important ethical principle is that experiments should include only procedures that people would agree to experience. No one should leave a study muttering, "If I had known what was going to happen, I never would have agreed to participate." To maintain high ethical standards for the conduct of experiments, psychologists ask prospective participants to give their **informed consent** before proceeding, ***a statement that they have been told what to expect and that they agree to continue.*** When experimenters post a sign-up sheet asking for volunteers, or at the start of the experiment itself, they explain what will happen—that the participants will receive electric shocks, or they will be asked to drink concentrated sugar water, or whatever. Any prospective participant who objects to the procedure can simply withdraw. Informed consent sounds like a simple principle, and usually it is. However, special problems arise with research on retarded or otherwise impaired people who may not understand the proposed procedure (Bonnie, 1997) and with severely depressed people, who sometimes agree to very risky experiments, partly because they have lost interest in their own well-being (Elliott, 1997). In such cases, it is sometimes necessary to consult the person's guardian or nearest relative, and sometimes wise investigators themselves decide not to proceed.

Experiments conducted at a college or at any other reputable institution must first be approved by a Human Subjects Committee at that institution. Such a committee judges whether the proposed studies include procedures for informed consent and whether they safeguard each participant's confidentiality. The committee also tries to prevent procedures that might expose participants to any serious risk. For example, the committee would not approve an experiment that called for administering large doses of cocaine, even if some of the subjects were eager to give their informed consent. The committee also judges procedures in which the investigators want to hide the purpose of the study. For example, suppose that a researcher is testing a method for changing people's attitudes. If the people knew that the researcher was trying to change their attitudes, they might resist strongly. It is also possible that they might pretend to change their attitudes, just to help the researcher. In either case, the results would be invalid. So the investigator might ask permission to conceal certain procedures, or even to mislead the participants temporarily, and the Human Subjects Committee would decide whether to permit the study under these conditions.

Finally, the American Psychological Association, or APA (1982), publishes a booklet detailing the proper ethical treatment of volunteers in experiments. Any member who disregards these principles may be censured or expelled from membership in the APA.

Ethical Concerns in Experiments on Animals

Much psychological research requires the use of human subjects—for example, research on reading or television viewing. However, animal research can help us understand basic processes, such as how nerves work, how the eyes and ears work, the functions of sleep, and the effects of rewards and punishments on the rates of behaviors (Figure 2.14). Researchers are especially likely to turn to animals if they want to control aspects of life that people will not let them control (such as who mates with whom), or if they want to study behavior continuously over months or years (longer than people are willing to stay in a laboratory), or if the research poses possible health risks. Animal research has long been essential for preliminary testing of most new drugs, surgical procedures, and methods of relieving pain. People with untreatable illnesses argue that they have the right "to hope for cures or relief from suffering through research using animals" (Feeney, 1987). Even outside the medical field, much of our current knowledge in psychology either began with animal research or made use of animal studies at some point.

Nevertheless, some people oppose much or all of animal research. Animals, after all, are unable to give informed consent. Some animal rights supporters insist that animals should have the same rights as humans, that keeping animals (even pets) in cages is nothing short of slavery, and that killing any animal is murder. Others oppose some kinds of research but are willing to compromise about others. Psychologists, too, vary in their attitudes. Most support animal research in general, but most would draw a line somewhere separating acceptable research from unacceptable (Plous, 1996). Naturally, psychologists draw that line at different places.

In this debate, as in so many other political controversies, one common tactic is for each side to criticize the most extreme actions of their opponents. For example, animal-rights advocates point to studies that exposed monkeys or puppies to painful procedures that seem hard to justify. On the other hand, researchers point to protesters who have distorted facts, vandalized laboratories, and even threatened to kill researchers and their children. Some protesters have even stated that they would oppose the use of any AIDS medication if its discovery came out of research with animals.

Unfortunately, when both sides concentrate on criticizing their most extreme opponents, they trivialize the real issues and make points of agreement more difficult to find. One fairly thorough study by a relatively unbiased outsider concluded that the truth is messy: Some research is painful or harmful to animals *and* nevertheless valuable for advancing our understanding of important scientific or medical issues in ways that we could not otherwise achieve (Blum, 1994). In other words, we cannot remain innocent of harming animals and still make the scientific and medical progress that we value.

If we cannot simply choose between progress and protecting animals, we must find a compromise. Professional organizations such as the Neuroscience Society and the American Psychological Association publish guidelines for the proper use of animals in research. Colleges and other research institutions maintain laboratory animal care committees to ensure that laboratory animals are treated humanely, that their pain and discomfort are kept to a minimum, and that experimenters consider alternatives before they impose potentially painful procedures on animals. Because such committees must deal with competing values, their decisions are never beyond dispute. How can we determine in advance whether the value of the expected

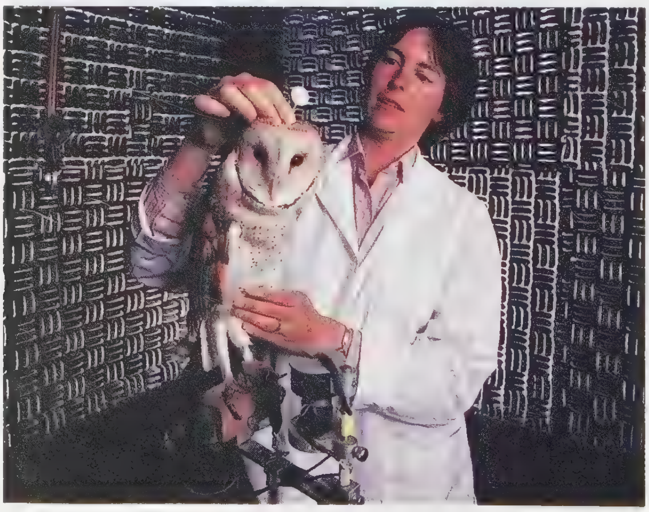


FIGURE 2.14 A mirror mounted on a young owl's head enables investigators to track the owl's head movements and thereby to discover how it localizes sounds with one ear plugged. The findings may help researchers understand how blind people use their hearing to compensate for visual loss. An experiment such as this subjects the animal to only a minor inconvenience. Some experiments, however, inflict pain or discomfort and are therefore more likely to raise ethical objections.

experimental results (which are hard to predict) will outweigh the pain the animals will endure (which is hard to measure)? As is so often the case with ethical decisions, reasonable arguments can be raised on both sides of the question, and no compromise is fully satisfactory.

THE MESSAGE
Psychological Research

I mentioned at the beginning of this chapter that scientists avoid the word *prove*. Psychologists certainly do. (The joke is that psychology courses do not have true-false tests, just maybe-perhaps tests.) The most complex, and therefore most interesting, aspects of human behavior are products of genetics, a lifetime of experiences, and countless current influences. Given the practical and ethical limitations, it might seem that the whole idea of psychological research is doomed from the start. However, largely because of these difficulties, many psychologists have been highly inventive in designing complex ways to isolate the effects of certain influences. A single study rarely answers a question decisively, but many studies can converge to increase our total understanding.

SUMMARY

✷ *Operational definitions.* For many purposes, psychologists prefer operational definitions, which state how to measure a given phenomenon or how to produce it. (page 37)

✷ *Sampling.* Psychologists hope to draw conclusions that apply to a large population and not just to the small sample they have studied, so they try to select a sample that resembles the total population. They may select either a representative sample or a random sample. To apply the results to people worldwide, they need a cross-cultural sample. (page 38)

✷ *Experimenter bias and blind observers.* An experimenter's expectations can influence the interpretations of behavior and the recording of data. To ensure objectivity, investigators use blind observers, who do not know what results are expected. In a double-blind study, neither the observer nor the subjects know the researcher's predictions. (page 39)

✷ *Naturalistic observations.* Naturalistic observations provide descriptions of humans or other species under natural conditions. (page 40)

✷ *Case histories.* A case history is a detailed research study of a single individual, generally someone with unusual characteristics. (page 40)

✷ *Surveys.* A survey is a report of people's answers on a questionnaire. It is fairly easy to conduct a survey and, unfortunately, very easy to conduct a survey badly. (page 41)

✷ *Correlations.* A correlational study is a study of the relationship between variables that are outside the investigator's control. The strength of this relationship is measured by a correlation coefficient, which ranges from 0 (no relationship) to plus or minus 1 (a perfect relationship). (page 43)

✷ *Illusory correlations.* Beware of illusory correlations—relationships that people think they observe between variables after mere casual observation. (page 44)

✷ *Inferring causation.* A correlational study will not uncover cause-and-effect relationships, but an experiment can. (page 44)

✷ *Experiments.* Experiments are studies in which the investigator manipulates one variable to determine its effect on another variable. The manipulated variable is the independent variable. Changes in the independent variable may lead to changes in the dependent variable, the one the experimenter measures. (page 45)

✷ *Random assignment.* An experimenter should randomly assign individuals to form experimental and control groups. That is, all individuals should have an equal probability of being chosen for the experimental group. (page 46)

✷ *Demand characteristics.* Cues that inform participants of the expected results are called demand characteristics. Skillful experimenters try to minimize demand characteristics. (page 48)

✷ *Ethics of experimentation.* Experimentation on either humans or animals raises ethical questions. Psychologists try to minimize risk to their subjects, but they cannot avoid making difficult ethical decisions. (page 49)

Suggestion for Further Reading

Stanovich, K. E. (1998). *How to think straight about psychology* (5th ed.). Reading, MA: Addison-Wesley. An excellent discussion of how to evaluate evidence in psychology and how to avoid pitfalls.

Terms

operational definition a definition that specifies the operations (or procedures) used to produce or measure something, a way to give it a numerical value (page 37)

convenience sample a group chosen because of its ease of study (page 38)

representative sample a selection of the population chosen to match the entire population with regard to specific variables (page 38)

random sample a group of people picked in random fashion, so that every individual in the population has an equal chance of being selected (page 38)

cross-cultural samples groups of people from at least two cultures (page 38)

experimenter bias the tendency of an experimenter to unintentionally distort procedures or results, based on the experimenter's own expectations of the outcome of the study (page 39)

blind observer an observer who can record data without knowing what the researcher has predicted (page 39)

placebo an inactive pill that has no known pharmacological effect on the subjects in an experiment (page 40)

single-blind study a study in which either the observer or the subjects are unaware of which subjects received which treatment (page 40)

double-blind study a study in which neither the observer nor the subjects know which subjects received which treatment (page 40)

naturalistic observation a careful examination of what many people or nonhuman animals do under natural conditions (page 40)

case history a thorough description of a single individual, including information on both past experiences and current behavior (page 41)

survey a study of the prevalence of certain beliefs, attitudes, or behaviors, based on people's responses to specific questions (page 41)

correlation a measure of the relationship between two variables, which are both outside the investigator's control (page 43)

correlation coefficient a mathematical estimate of the relationship between two variables, ranging from +1 (perfect positive relationship) to 0 (no linear relationship) to −1 (perfect negative relationship) (page 43)

illusory correlation an apparent relationship based on casual observations of unrelated or weakly related events (page 44)

experiment a study in which the investigator manipulates at least one variable while measuring at least one other variable (page 45)

independent variable in an experiment, the item that an experimenter manipulates to determine how it affects the dependent variable (page 45)

dependent variable the item that an experimenter measures to determine how changes in the independent variable affect it (page 45)

experimental group the group that receives the treatment that an experiment is designed to test (page 46)

control group a group treated in the same way as the experimental group except for the procedure that the experiment is designed to test (page 46)

random assignment a chance procedure for assigning subjects to groups so that every subject has the same probability as any other subject of being assigned to a particular group (page 46)

demand characteristics cues that tell a subject what is expected of him or her and what the experimenter hopes to find (page 49)

informed consent a subject's agreement to take part in an experiment after being told what to expect (page 49)

Answers to Concept Checks

3. A score on an IQ test is an operational definition of intelligence. (Whether it is a particularly good definition is another question.) None of the other definitions tells us how to measure or produce intelligence. Many operational definitions are possible for "friendliness," such as "the number of people that someone speaks to within 24 hours." You can probably think of a better operational definition. Remember, that an operational definition specifies a clear method of measurement. (page 38)
4. Clearly not. It is unlikely that the men at a given college are typical of men in general or that the women are typical of women in general. Moreover, at some colleges the men are atypical in some respects and the women atypical in different ways. (page 39)
5. The −0.75 correlation indicates a stronger relationship—that is, a greater accuracy of predicting one variable based on measurements of the other. A negative correlation is just as useful as a positive one. (page 44)
6. We can conclude only that, if we know either someone's interest level or test score, we can predict the other with reasonably high accuracy. We *cannot* conclude that an interest in psychology will help someone to learn the material or that doing well on psychology tests increases someone's interest in the material. Both conclusions may well be true, but a correlational study cannot demonstrate a cause-and-effect relationship. (page 45)
7. The independent variable is the frequency of tests during the semester. The dependent variable is the students' performance on the final exam. (page 46)
8. Of these, only demand characteristics are to be avoided. If you did not remember that falsifiability is a good feature of a theory, check page 31. Every experiment must have at least one independent variable (what the experimenter controls) and at least one dependent variable (what the experimenter measures). Blind observers provide an advantage. (page 49)

Web Resources

HyperText Psychology—BASICS
www.science.wayne.edu/~wpoff/basics.html
This site walks you through the basics of research: social science methods; field studies, surveys, and experiments; and the use of statistics in interpreting results.

MODULE 2.3

Measuring and Analyzing Results

How can the "average" results in a study be described?

How can we describe the variations among individuals?

How can a researcher determine whether the results reflect a consistent trend, or whether these results could have arisen just by chance?

Some years ago, a television program about the alleged dangers of playing the game Dungeons and Dragons reported 28 known cases of D&D players who had committed suicide. Alarming, right?

Not necessarily. At that time at least 3 million young people played the game regularly. The reported suicide rate among D&D players—28 per 3 million—was considerably *less* than the suicide rate among teenagers in general.

So, do these results mean that playing D&D *prevents* suicide? Hardly. The 28 reported cases are probably an incomplete count of all suicides by D&D players. Besides, the correlation between playing D&D and committing suicide, regardless of its direction and magnitude, could not possibly tell us about cause and effect. Maybe the kinds of young people who play D&D are simply different from those who do not.

Then, what conclusion should we draw from these data? *None at all.* Sometimes, as in this case, the data are meaningless because of how they were collected. Even when the data are potentially meaningful, people sometimes present them in a confusing or misleading manner (Figure 2.15). Let's consider some proper ways of analyzing and interpreting results.

Descriptive Statistics

To explain the meaning of a study, an investigator must summarize the results in an orderly fashion. When a researcher observes the behavior of 100 people, we have no interest in hearing all the details about every person observed. We want to know what the researcher found in general, on the average. We might also want to know whether most people were similar to the average or whether they varied a great deal. An investigator presents the answers to those questions through **descriptive statistics,** which are ***mathematical summaries of results,*** such as measures of the average and the amount of variation. The correlation coefficient, discussed earlier in this chapter, is one kind of descriptive statistic.

Measurements of the Central Score: Mean, Median, and Mode

There are three ways of representing the central score: mean, median, and mode. The **mean** is ***the sum of all the scores divided by the total number of scores.*** (Generally,

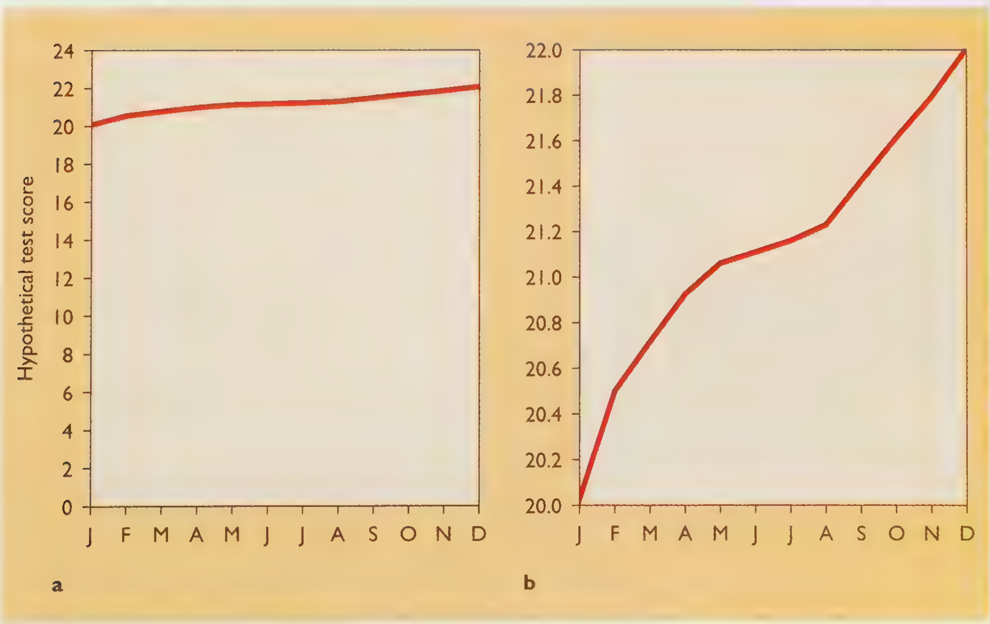


FIGURE 2.15 Why statistics can be misleading: Both of these graphs present the same data, an increase from 20 to 22 over one year's time. But by ranging only from 20 to 22 (rather than from 0 to 22), graph (b) makes that increase look much more dramatic. (After Huff, 1954.)

when people say "average," they refer to the mean.) For example, the mean of 2, 10, and 3 is 5 (15 ÷ 3). The mean is especially useful if the scores approximate the **normal distribution** (or normal curve), ***a symmetrical frequency of scores clustered around the mean.*** A normal distribution is sometimes described as a bell-shaped curve. For example, if we measure how long it takes 30 students to memorize a poem, these times will probably follow a pattern similar to the normal distribution.

The mean can be misleading, however, if the distribution is far from normal. Suppose, for example, we want to find out whether an article published in 1998 relied on up-to-date information. We check the dates of the references cited in the article and find the distribution shown in Figure 2.16. Here, the scores follow an approximately normal distribution, and the mean of 1994 adequately represents the results. However, suppose the author had added three quotes—one from Shakespeare (published in 1610), one from Aristotle (324 B.C.), and one from *Genesis* (4000? B.C.). The addition of those three references would lower the mean date from 1994 to 1841, giving the very misleading impression that the article had relied on out-of-date information.

The addition of these three old references would have much less effect on the median, however. To determine the **median,** ***we arrange all the scores in order from the highest score to the lowest score. The middle score is the median.*** For example, if the scores are 2, 10, and 3, the median is 3. In Figure 2.16 the median date of the references is 1994. After adding three very old references, the median is still 1994. If the author had added eight more old references, the median would fall to 1993. In short, a few extreme scores have less effect on the median than they do on the mean.

The third way to represent the central score is the **mode,** ***the score that occurs most frequently.*** For example, in the distribution of scores 2, 2, 3, 4, and 10, the mode is 2.

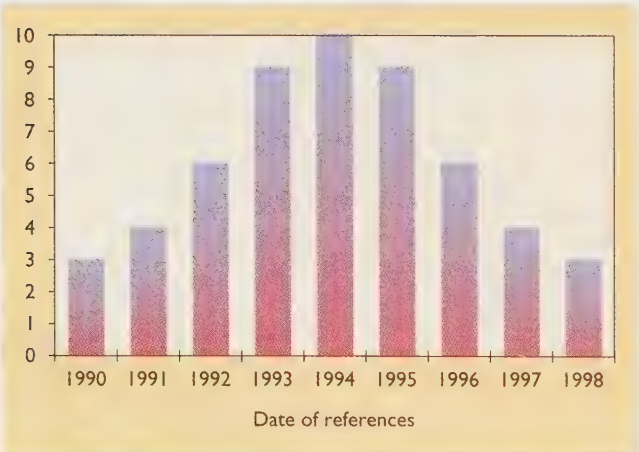


FIGURE 2.16 A distribution of references for a hypothetical publication: Note that the distribution is symmetrical and similar to the normal curve. In this case, the mean, median, and mode are all the same: 1994.

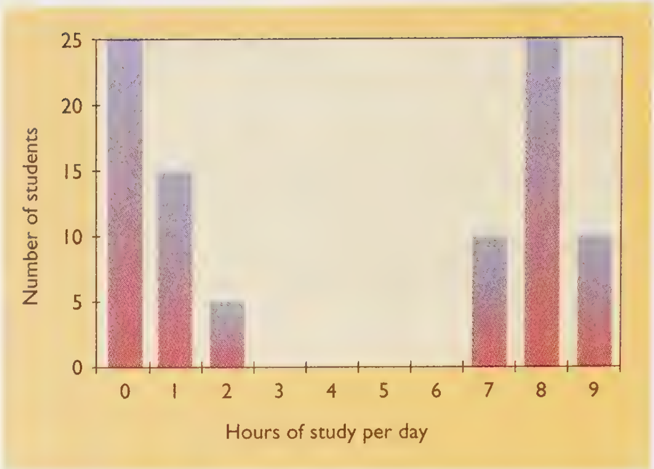


FIGURE 2.17 Results of an imaginary survey of study habits at one college. This college apparently has two groups of students—those who study as hard as they can, and those who find other things to do. In this case, both the mean and the median are misleading. This distribution is bimodal; its two modes are 0 and 8.

The mode is less useful than the mean or median, except under special circumstances. Suppose we surveyed a sample of students at a college about how much they study and gathered the results shown in Figure 2.17. Half of the students study a great deal, and half of them study very little. The mean for this distribution is 4.28 hours per day, a very misleading figure because all the students study either much more or much less than that. In this case, the median is no better as a representation of the results: Because we have an even number of students, there is no middle score. We could take a figure midway between the two scores nearest the middle, but in this case those scores are 2 and 7, so we would compute a median of 4.5, again a very misleading figure. A distribution like this is called a *bimodal distribution* (one with two modes); the researcher might simply describe the two modes and not even mention the mean or the median.

To summarize: Roughly speaking, the mean is what most people intend when they say "average." The median is the middle score after the scores are ranked from highest to lowest; the mode is the most common score (Figure 2.18).

CONCEPT CHECK

9. a. For the following distribution of scores, determine the mean, the median, and the mode: 5, 2, 2, 2, 8, 3, 1, 6, 7.

b. Determine the mean, median, and mode for this distribution: 5, 2, 2, 2, 35, 3, 1, 6, 7. (Check your answers on page 58.)

10. A survey of college men asked, "Ideally, how many sexual partners would you like to have in the rest of your life?" The mean answer was 64. However, the median was 1. How can we explain this discrepancy? (Check your answers on page 58.)

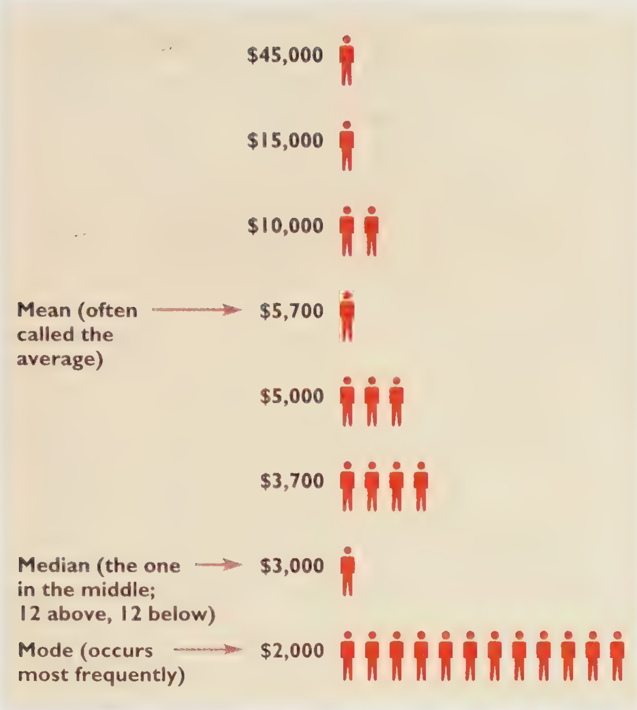


FIGURE 2.18 The monthly salaries of the 25 employees of company X, showing the mean, median, and mode. (After Huff, 1954.)

Measures of Variation

Figure 2.19 shows two distributions of test scores. Suppose that these represent scores on two introductory psychology tests. Both tests have the same mean, 70, but different distributions. If you had a score of 80, you would beat only three fourths of the other students on the first test, but with the same score you would beat 95% on the second test.

To describe the difference between Figure 2.20a and b, we need a measurement of the variation (or spread) around the mean. The simplest such measurement is the **range** of a distribution, ***a statement of the highest and lowest scores.*** The range in Figure 2.20a is 39 to 100, and in Figure 2.20b it is 58 to 92.

The range is a simple calculation, but it is not very useful, because it reflects only the extremes. Statisticians need to know whether most of the scores are clustered close to the mean or scattered widely. The most useful measure is the **standard deviation (SD),** ***a measurement of the amount of variation among scores in a normal distribution.*** In the appendix to this chapter, you will find a formula for calculating the standard deviation. For present purposes, you can simply remember that when the scores are closely clustered near the mean, the standard deviation is small; when the scores are more widely scattered, the standard deviation is large.

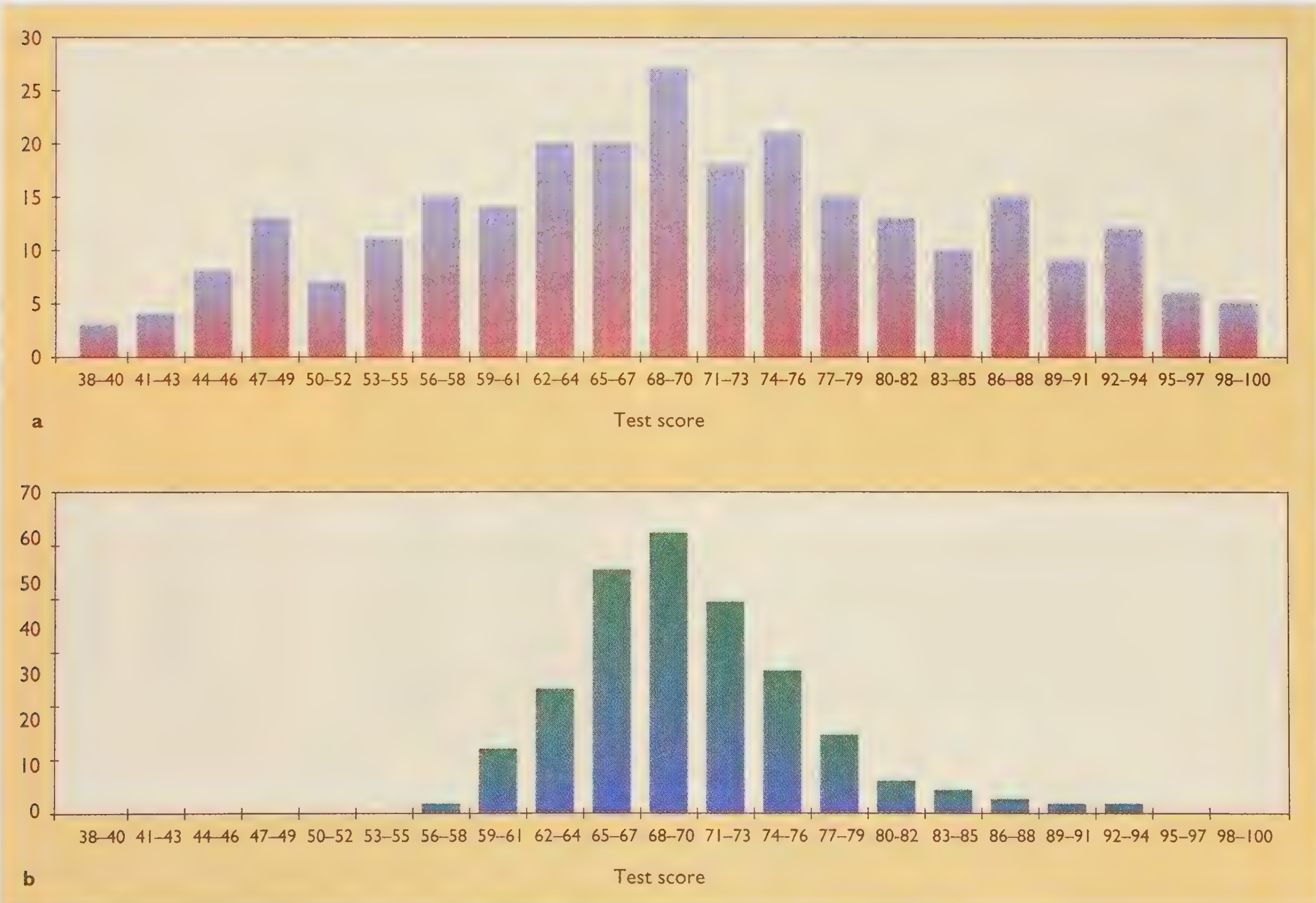


FIGURE 2.19 These two distributions of test scores have the same mean but different variances and different standard deviations.

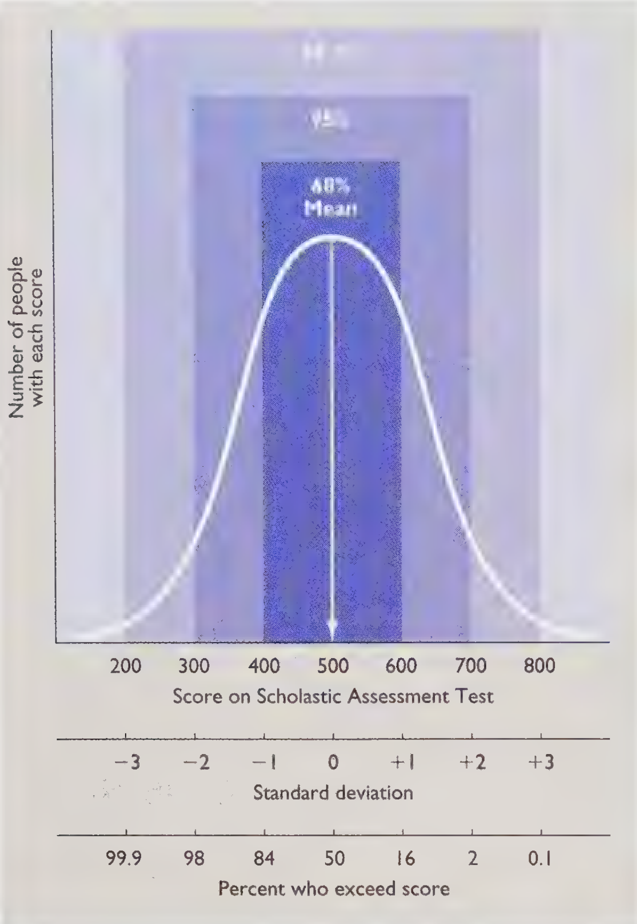


FIGURE 2.20 In a normal distribution of scores, the amount of variation from the mean can be measured in standard deviations. In this example, scores between 400 and 600 are said to be within one standard deviation from the mean; scores between 300 and 700 are within two standard deviations.

As Figure 2.20 shows, the Scholastic Assessment Test (SAT) was designed to produce a mean of 500 and a standard deviation of 100. Of all people taking the test, 68% score within one standard deviation above or below the mean (400–600); 95% score within two standard deviations (300–700). Only 2.5% score above 700; another 2.5% score below 300.

Standard deviations provide a useful way of comparing scores on different tests. For example, if you scored one standard deviation above the mean on the SAT, you tested about as well as someone who scored one standard deviation above the mean on another test, such as the American College Test. We would say that both of you had a deviation score of +1.

CONCEPT CHECK

11. Suppose that you score 80 your first psychology test. The mean for the class is 70, and the standard deviation is 5. On the second test, you receive a score of 90. This time the mean for the class is also 70, but the standard deviation is 20. Compared to the other students in your class, did your performance improve, deteriorate, or stay the same? (Check your answer on page 58.)

Evaluating Results: Inferential Statistics

Suppose researchers conducted a study comparing two kinds of therapy for helping people to quit smoking cigarettes. At the end of six weeks of therapy, people who have been punished for smoking average 7.5 cigarettes per day, whereas those who have been rewarded for *not* smoking average 6.5 cigarettes per day. Presuming that the smokers were randomly assigned to the two groups and that the study was properly conducted, is this the kind of difference that might easily arise by chance? Or should we take this difference seriously and recommend that all therapists use rewards and not punishment?

To answer this question, we obviously need to know more than just the numbers 7.5 and 6.5. How many smokers were in the study? (10 in each group? a hundred? a thousand?) And how much variation occurred within each group? Are most people's behaviors close to the group means, or are there a few extreme scores that distort the averages?

One way to deal with these issues is to present the means and 95% confidence intervals for each, as shown in Figure 2.21. The **95% confidence interval** is ***the range within which the true population mean lies, with 95% certainty.*** "Wait a minute," you protest. "We already know the means: 7.5 and 6.5. Aren't those the *true* population means?" No, those are the means for particular samples of the population. Someone who studies another group of

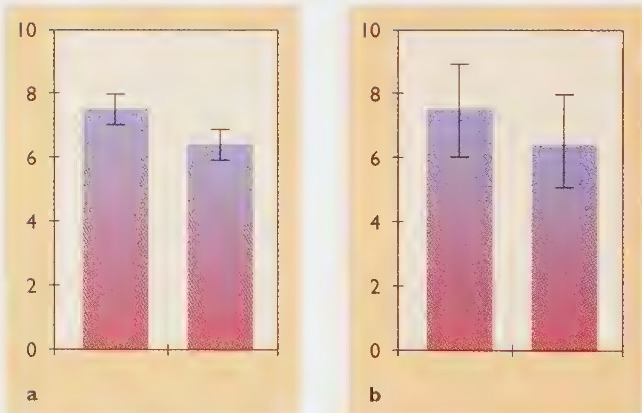


FIGURE 2.21 The vertical lines indicate 95% confidence intervals. The pair of graphs in part (a) indicate that the true mean has a 95% chance of falling within a very narrow range. The graphs in (b) indicate a wider range and therefore suggest less certainty that reward is a more effective therapy than punishment.